

Signed and Reciprocal q -Fabius Frontiers

Negative parameters, reciprocal Laplace duality, two-nome
Pochhammer–Prouhet partition functions, and inverse-geometric endpoint lattices

A frontier research report based on the recursive `*.tex` corpus under
Analysis/FabiusFunction/docs

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Abstract

The existing PROVEIt corpus develops the classical Fabius distribution and Rvachev up-function at dyadic scale, exact dyadic arithmetic, the Thue–Morse finite-block calculus, half-base Gaussian coefficients, q -Pochhammer products, finite sinc products, Bell–Bernoulli endpoint expansions, Lambert- W_{-1} saddle coordinates, and several $q = 1/4$ Richardson schemes. The purpose of this report is not to repackage that material. It studies instead the signed and reciprocal parameter orbit

$$\{q, -q, q^{-1}, -q^{-1}\}$$

after separating three roles that are frequently denoted by the same letter: the geometric ratio of the random series, the Gaussian-binomial nome, and the digit-position weight in a Thue–Morse projection.

For $|q| < 1$ define the normalized geometric-uniform variable

$$Y_q = (1 - q) \sum_{j \geq 0} q^j U_j, \quad U_j \sim \text{Unif}[0, 1]$$

independently, and let $M_q(t) = \mathbb{E}e^{tY_q}$. The case $q = 1/2$ is exactly the Fabius distribution. We extend M_q to $|q| > 1$ as a normalized meromorphic moment germ satisfying the same Mahler equation. The resulting system obeys

$$M_q(t)M_{1/q}(-t) = 1, \quad Y_{-a} \stackrel{d}{=} \frac{1+a}{1-a}Y_a - \frac{a}{1-a} \quad (0 < a < 1).$$

Thus the requested values split into two Klein-four orbits, $\{\pm \frac{1}{2}, \pm 2\}$ and $\{\pm \frac{1}{4}, \pm 4\}$, and every member is obtained from one positive contraction by affine transport and reciprocal inversion. The spectral product has nome q^2 : sign change rescales the centered germ, whereas inversion exchanges its zeros and poles.

Several further consequences are proved. First, the reciprocal centered germ is a genuine characteristic function: for $|q| > 1$ it is the Fourier transform of an explicit infinite convolution of Laplace laws. Hence the formal moments are moments of a positive measure on the vertical line $\Re z = 1/2$, their Hankel determinants have exact sign $(-1)^{\binom{n}{2}}$, and their monic orthogonal polynomials have all zeros on that line. Second, the inverse averaging operator produces a q -Fabius–Bernoulli Appell family interpolating ordinary Bernoulli polynomials at $q = 0$ and centered monomials as $q \rightarrow 1^-$. A new polynomial

$$\mathcal{P}_n(q) = \frac{(q; q)_n}{(1 - q)^n} \frac{\mathbb{E}Y_q^n}{n!}$$

packages the moments and yields an exact q -Pochhammer formula for the geometric endpoint lattice.

The Thue–Morse and Gaussian layers are unified by the two-nome binary-cube partition function

$$\mathcal{Z}_N(u, t; \rho, \sigma) = \prod_{j=0}^{N-1} \left(1 - u\rho^j e^{t\sigma^j}\right).$$

Its $t = 0$ face is finite q -Pochhammer/Gaussian-binomial calculus; its $(u, \rho) = (1, 1)$ face is a geometric Prouhet generator. From it we obtain an exact q -Prouhet moment-transfer theorem, signed position-energy layers, simultaneous multisection in both nomes, and a finite-field/unitary square for $q = \pm 1/2, \pm 2, \pm 1/4, \pm 4$. The associated box-spline derivative comb is the full dyadic Thue–Morse grid at $q = 1/2$ but a signed Cantor skeleton of Hausdorff dimension $1/2$ at $q = 1/4$.

Finally, for $0 < q \leq 1/2$ we prove the inverse-geometric endpoint identity

$$G_q(q^n) = \frac{q^{\binom{n+1}{2}}}{(1-q)^n} \frac{\mathbb{E}Y_q^n}{n!} = \frac{q^{\binom{n+1}{2}}}{(q; q)_n} \mathcal{P}_n(q),$$

including all derivative jets there. It gives an exact inverse-quartic lattice for $q = 1/4$ and yields the uniform two-term endpoint and inverse asymptotics

$$\log G_q(x) = -\frac{\log^2(1/x)}{2 \log(1/q)} - \frac{\log(1/x) \log \log(1/x)}{\log(1/q)} + O(\log(1/x)),$$

$$\log \frac{1}{G_q^{-1}(y)} = \sqrt{2 \log(1/q) \log(1/y)} - \frac{1}{2} \log \log(1/y) + O(1).$$

A generalized Lambert- W_{-1} all-orders program, natural-boundary conjectures, arithmetic questions for $\mathcal{P}_n$, and a Lean formalization roadmap conclude the report. A fully commented Python program verifies the principal orbit, Prouhet, endpoint, determinant, and moment-polynomial identities at the eight requested parameter values and supplies the figures and CSV tables in the accompanying archive.

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1 Audit boundary, terminology, and principal results

1.1 Pinned corpus and novelty discipline

The source boundary for this report is commit 89b1a5f915cad065a42ee48f758eb2cd2d9beb94. The repository was changing on the audit date, so every statement of novelty is relative to this exact snapshot. The audit followed the repository's own stratification:

- the proof-backed primary exposition and its Lean coverage map;
- the canonical non-formalized frontier volume;
- the thematic synthesis volumes for Thue–Morse, exponent sequences and q -series, spectra and arithmetic, representations, inverse sampling, and integration/transforms;
- archived q -calculus and dyadic-connection reports, treated as mathematical lineage rather than independent current claims;
- targeted code and formula searches across the recursive documentation tree for the identities proposed below.

The most important prior-art boundary is the consolidated volume `non-formalized-research-frontiers/drafts/exponents-and-q-series/Exponents_and_q_Series_Frontiers/Exponents_and_q_Series_Frontiers.tex`. It already contains arbitrary nonnegative exponent-sequence sinc products, zero multiplicities, spectral zeta and cumulant dictionaries, Gaussian q -binomial tomography, the Pascal–Rvachev hierarchy, $q = 1/4$ Richardson acceleration, finite-product spline approximants, sharp transport laws, and Mellin–Lambert endpoint mechanisms. The Thue–Morse atlas already contains the standard product, Prouhet cancellation, Walsh/Fourier formulae, automatic and rarefied sums, and the bridge to the Fabius/Rvachev system. None of those is claimed as new here.

Targeted searches in the pinned tree found no exact counterpart of the following package:

1. the normalized signed/reciprocal moment germ on the full orbit $\{q, -q, q^{-1}, -q^{-1}\}$ and its Cayley-coordinate action;
2. the affine conjugacy of negative geometric-uniform laws;
3. the reciprocal Laplace-convolution representation, exact Hankel signature, and vertical-line orthogonality;
4. the two-nome partition function combining Gaussian position energy and geometric Prouhet projection;
5. the all- q finite Prouhet moment-transfer identity;
6. the inverse-geometric endpoint formula, its derivative jets, and its $q = 1/4$ specialization;
7. the q -Fibius–Bernoulli inverse-averaging Appell family;
8. the moment polynomial $P_n(q)$ and its endpoint q -Pochhammer form.

External searches located the negative- q Gaussian-binomial interpretation of Fu, Reiner, Stanton, and Thiem, but not the Fabius/Rvachev reciprocal and endpoint structures assembled here. Accordingly, **new relative to the pinned snapshot** means only “not found in the pinned corpus after targeted audit and not found as a named package in the targeted literature search.” It is not a claim of exhaustive worldwide priority.

Proof-status convention. Every theorem marked **proved here** has a proof in this manuscript. The label **derived from corpus infrastructure** means that the argument uses an established object or theorem from the audited corpus and adds a new deduction. Statements marked **conjectural/open** are explicitly conjectural. Numerical evidence never replaces a proof; its role is reproducibility, error detection, and conjecture triage.

1.2 Three nomes that should not be conflated

The letter q occurs in several mathematically different roles. We keep them separate whenever two of them coexist:

symbol	role	typical object
q	geometric ratio	$Y_q = (1 - q) \sum q^j U_j, M_q, \Phi_q$
ρ	position-energy/Gaussian nome	$(u; \rho)_N, \left[\begin{smallmatrix} N \\ k \end{smallmatrix} \right]_\rho$
σ	geometric digit projection	$A_{\sigma, N}(\epsilon) = \sum \epsilon_j \sigma^j$
Q	finite-field cardinality	$\mathbb{F}_Q$, Hermitian space over $\mathbb{F}_{Q^2}$
q^2	spectral Pochhammer nome	zeros/poles of the centered sinc/hyperbolic product

The classical Fabius case sets $q = 1/2$. A finite Gaussian stencil derived from that case often also has nome $\rho = 1/2$, but this equality is contingent, not structural. The master partition function in [section 10](#) allows ρ and σ to vary independently and thereby exposes which statement belongs to which layer.

1.3 The two requested Klein-four orbits

The transformations $s(q) = -q$ and $r(q) = q^{-1}$ commute and generate a Klein four-group away from $q = 0$. The values requested in the problem are exactly two orbits:

$$\mathcal{O}_2 = \left\{ \frac{1}{2}, -\frac{1}{2}, 2, -2 \right\}, \quad \mathcal{O}_4 = \left\{ \frac{1}{4}, -\frac{1}{4}, 4, -4 \right\}.$$

The following table previews their formal means and variances. The word “formal” is needed only outside the unit disk; inside it these are ordinary probability moments.

Table 1: The requested signed/reciprocal parameter orbits.

q	geometric status	mean	variance/formal second cumulant
$1/2$	Fabius probability law	$1/2$	$1/36$
$-1/2$	affine Fabius law on $[-1, 2]$	$1/2$	$1/4$
2	reciprocal moment germ	$1/2$	$-1/36$
-2	reciprocal signed germ	$1/2$	$-1/4$
$1/4$	quartic geometric probability law	$1/2$	$1/20$
$-1/4$	affine quartic law on $[-1/3, 4/3]$	$1/2$	$5/36$
4	reciprocal quartic germ	$1/2$	$-1/20$
-4	reciprocal signed quartic germ	$1/2$	$-5/36$

1.4 Executive theorem map

The report has four mutually reinforcing layers.

1. **Moment/Mahler layer.** A single functional equation defines a probability MGF for $|q| < 1$ and a reciprocal germ for $|q| > 1$. Bell and Bernoulli polynomials give every coefficient.
2. **Spectral layer.** The centered transform is an infinite q^2 -Pochhammer product. Sign change is an affine rescaling; inversion exchanges zeros and poles and produces a positive Laplace spectral dual.
3. **Binary-cube layer.** A two-nome partition function contains Gaussian coefficients, signed position-energy layers, Thue–Morse Prouhet cancellation, and multisection as different specializations.
4. **Endpoint layer.** A simplex-volume argument transfers the moment system to an exact geometric endpoint lattice and its derivative jets, with a uniform first asymptotic inversion and a conjectural all-orders Lambert- W_{-1} continuation.

2 The normalized geometric-uniform law and its reciprocal germ

2.1 The positive and signed probability regimes

Let $U_0, U_1, \dots$ be independent $\text{Unif}[0, 1]$ random variables. For a real or complex parameter q with $|q| < 1$, define

$$Y_q = (1 - q) \sum_{j=0}^{\infty} q^j U_j. \quad (2.1)$$

The series converges absolutely, and its weights sum to one. Consequently

$$\mathbb{E}Y_q = \frac{1}{2} \quad (2.2)$$

for every real $|q| < 1$, including negative q .

For $0 \leq q < 1$, all weights are nonnegative and the support is $[0, 1]$. The case $q = 1/2$ is the random-series representation of the classical bounded Fabius function. We write

$$G_q(x) = \mathbb{P}(Y_q \leq x) \quad (2.3)$$

for the CDF in the positive regime. The existing Lean API in `GeometricUniformLaw.lean` formalizes the normalized weights, the affine self-similarity, reflection, absolute continuity, null singletons, and support.

For $q = -a$, $0 < a < 1$, positive and negative coefficient intervals add to

$$\sum_{j \text{ even}} (1 + a)a^j = \frac{1}{1 - a}, \quad - \sum_{j \text{ odd}} (1 + a)a^j = -\frac{a}{1 - a}.$$

Hence

$$\text{supp}(Y_{-a}) = \left[-\frac{a}{1 - a}, \frac{1}{1 - a} \right]. \quad (2.4)$$

The support remains an interval, not a Cantor set, because each coordinate is itself an interval variable.

2.2 The universal Mahler equation

Put

$$E(z) = \frac{e^z - 1}{z}, \quad E(0) = 1, \quad S(z) = e^{-z/2} E(z) = \frac{\sinh(z/2)}{z/2}. \quad (2.5)$$

For $|q| < 1$, independence gives the entire moment generating function

$$M_q(t) = \mathbb{E}e^{tY_q} = \prod_{j=0}^{\infty} E((1 - q)q^j t). \quad (2.6)$$

Splitting the factor $j = 0$ gives

$$M_q(t) = E((1 - q)t)M_q(qt). \quad (2.7)$$

This is the normalized geometric analogue of the classical Fabius moment Mahler equation.

Definition 2.1 (Reciprocal normalized germ). For $|q| > 1$, define near $t = 0$

$$M_q(t) = \left[\prod_{j=1}^{\infty} E((1 - q)q^{-j}t) \right]^{-1}. \quad (2.8)$$

It is meromorphic globally and analytic at the origin. Its Taylor coefficients are called the *reciprocal q -Fabius moments*.

The indexing in (2.8) is forced by normalization. Reindexing immediately proves that the same equation (2.7) holds for $|q| > 1$. Thus the unit circle, rather than the sign of q , is the natural dividing curve between the probability and reciprocal-germ regimes.

Theorem 2.2 (Uniqueness of the normalized formal solution). *Let $q \in \mathbb{C}$ satisfy $q^n \neq 1$ for every $n \geq 1$. There is a unique formal power series $M_q(t) \in \mathbb{C}[[t]]$ with $M_q(0) = 1$ satisfying (2.7). It is represented by (2.6) when $|q| < 1$ and by (2.8) when $|q| > 1$. This statement is [proved here](#).*

Proof. Write $M_q(t) = \sum_{n \geq 0} m_n(q)t^n/n!$. Comparing the coefficient of t^n in (2.7) gives

$$(1 - q^n)m_n(q) = \sum_{k=0}^{n-1} \binom{n}{k} \frac{(1 - q)^{n-k} q^k}{n - k + 1} m_k(q). \quad (2.9)$$

The coefficient $1 - q^n$ is nonzero, so m_n is determined recursively by $m_0 = 1$. The convergent products satisfy the same equation and normalization, therefore realize the unique formal solution. $\square$

For exponential coefficients $a_n(q) = m_n(q)/n!$, the recurrence is

$$(1 - q^n)a_n(q) = \sum_{r=1}^n \frac{(1 - q)^r q^{n-r}}{(r + 1)!} a_{n-r}(q). \quad (2.10)$$

It specializes at $q = 1/2$ to the half-moment recurrence used throughout the classical Fabius literature.

2.3 Reflection and centered normalization

Since $E(-z) = e^{-z}E(z)$ and the normalized weights sum to one, both convergent products and the formal solution satisfy

$$M_q(-t) = e^{-t}M_q(t). \quad (2.11)$$

Define the centered germ

$$C_q(t) = e^{-t/2}M_q(t). \quad (2.12)$$

Then

$$C_q(-t) = C_q(t), \quad (2.13)$$

so every centered odd cumulant vanishes. In the probability regime this is ordinary reflection of Y_q about $1/2$; outside the unit disk it remains an exact algebraic symmetry of the germ.

3 Bernoulli cumulants and Bell-polynomial moments

3.1 All cumulants in one rational function of q

The expansion

$$\log E(z) = \frac{z}{2} + \sum_{n=2}^{\infty} \frac{B_n^+}{n} \frac{z^n}{n!} \quad (3.1)$$

uses the Bernoulli convention $B_1^+ = +1/2$; for $n \neq 1$ these are the ordinary Bernoulli numbers. Summing over the geometric scales gives the following closed form.

Theorem 3.1 (Signed/reciprocal cumulant dictionary). *For every nonsingular q ,*

$$\boxed{\kappa_1(q) = \frac{1}{2}, \quad \kappa_n(q) = \frac{B_n^+}{n} \frac{(1-q)^n}{1-q^n} \quad (n \geq 2).} \quad (3.2)$$

In particular, $\kappa_{2m+1} = 0$ for $m \geq 1$ and

$$\kappa_2(q) = \frac{1}{12} \frac{1-q}{1+q}. \quad (3.3)$$

The theorem is [proved here](#).

Proof. For $|q| < 1$, insert (3.1) into (2.6) and sum $\sum_{j \geq 0} q^{jn} = (1 - q^n)^{-1}$. Both sides of (3.2) are rational functions of q coefficient by coefficient and satisfy the unique formal recurrence of [theorem 2.2](#); equivalently one may repeat the calculation in the reciprocal product. The variance formula follows from $B_2 = 1/6$. $\square$

The simple Möbius function $(1 - q)/(1 + q)$ already predicts the sign pattern in [table 1](#): inversion changes the sign of every even centered cumulant,

$$\kappa_{2m}(1/q) = -\kappa_{2m}(q). \quad (3.4)$$

Sign change rescales them by an even power, as proved in [section 4](#).

3.2 Complete Bell-polynomial reconstruction

Let $\mathcal{B}_n(x_1, \dots, x_n)$ denote the complete exponential Bell polynomial, characterized by

$$\exp\left(\sum_{j \geq 1} x_j \frac{t^j}{j!}\right) = \sum_{n \geq 0} \mathcal{B}_n(x_1, \dots, x_n) \frac{t^n}{n!}.$$

Then

$$\boxed{m_n(q) = \mathcal{B}_n\left(\frac{1}{2}, \kappa_2(q), 0, \kappa_4(q), 0, \dots\right).} \quad (3.5)$$

This makes all requested parameter values exactly computable over $\mathbb{Q}$. The first moments are

$$\begin{aligned} m_0 &= 1, & m_1 &= \frac{1}{2}, \\ m_2 &= \frac{1}{4} + \kappa_2, & m_3 &= \frac{1}{8} + \frac{3}{2}\kappa_2, \\ m_4 &= \frac{1}{16} + \frac{3}{2}\kappa_2 + 3\kappa_2^2 + \kappa_4. \end{aligned} \quad (3.6)$$

The accompanying CSV file `data/q_orbit_moments.csv` lists m_0 through m_6 for all eight requested values.

Remark 3.2 (Why the reciprocal moments are not probability moments). For $|q| > 1$, (3.3) is negative whenever $q > 1$. Therefore M_q cannot be the MGF of a real probability measure. This obstruction is not a defect; it points to the vertical-line and Laplace representation in [section 7](#), where the same numbers become honest complex moments of a positive real measure.

4 The Klein-four sign/inversion calculus

4.1 Negative parameters as affine positive laws

Write $V_j = U_j - 1/2$. The variables V_j are independent and symmetric, so changing any subset of their signs does not change their joint law. Since

$$Y_q - \frac{1}{2} = (1 - q) \sum_{j \geq 0} q^j V_j, \quad (4.1)$$

negative geometric ratios are affine images of positive ones.

Theorem 4.1 (Negative- q affine conjugacy). *For $0 < a < 1$, put*

$$\lambda_a = \frac{1 + a}{1 - a}, \quad \beta_a = \frac{a}{1 - a}. \quad (4.2)$$

Then

$$\boxed{Y_{-a} \stackrel{d}{=} \lambda_a Y_a - \beta_a,} \quad \boxed{Y_{-a} - \frac{1}{2} \stackrel{d}{=} \lambda_a \left(Y_a - \frac{1}{2} \right).} \quad (4.3)$$

Equivalently,

$$M_{-a}(t) = e^{-\beta_a t} M_a(\lambda_a t), \quad C_{-a}(t) = C_a(\lambda_a t). \quad (4.4)$$

This theorem is *proved here* and *new relative to the pinned snapshot*.

Proof. In (4.1), replace $(-1)^j V_j$ by V_j in law:

$$Y_{-a} - \frac{1}{2} = (1 + a) \sum_{j \geq 0} (-a)^j V_j \stackrel{d}{=} (1 + a) \sum_{j \geq 0} a^j V_j = \lambda_a \left(Y_a - \frac{1}{2} \right).$$

Expanding the centered affine map gives the raw shift $-\beta_a$. Taking MGFs proves (4.4). $\square$

The two examples most relevant here are

$$Y_{-1/2} \stackrel{d}{=} 3Y_{1/2} - 1, \quad Y_{-1/4} \stackrel{d}{=} \frac{5}{3}Y_{1/4} - \frac{1}{3}. \quad (4.5)$$

Thus the negative-half law is not a new shape: it is the classical Fabius density stretched by three and translated to $[-1, 2]$. The negative-quarter law is the quartic density stretched by $5/3$ and translated to $[-1/3, 4/3]$.

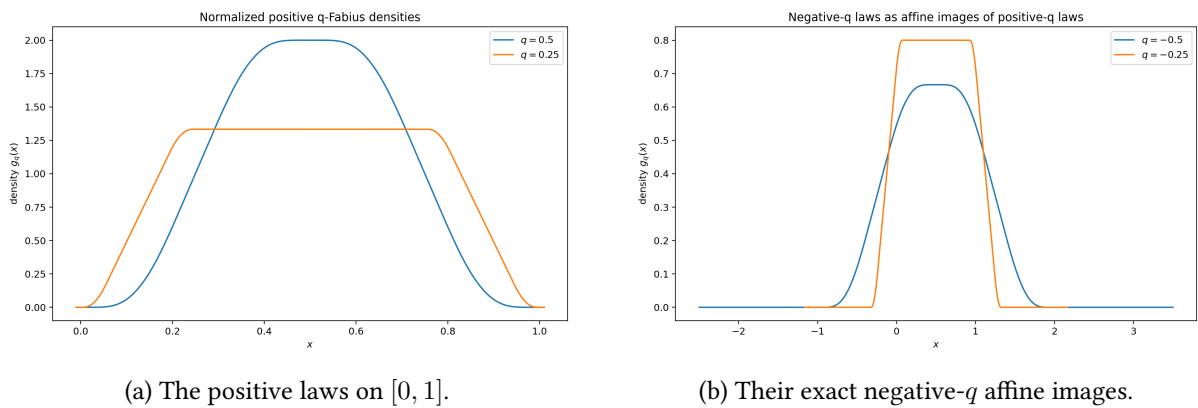


Figure 1: Deterministic Fourier inversion of finite characteristic products. The second panel contains no independent numerical solve: it is obtained from [equation \(4.3\)](#).

4.2 Reciprocal inversion

Theorem 4.2 (Reciprocal convolution inversion). *For every q off the unit circle,*

$$\boxed{M_q(t)M_{1/q}(-t) = 1,} \quad \boxed{C_q(t)C_{1/q}(t) = 1.} \quad (4.6)$$

*Both equalities hold as analytic germs at $t = 0$ and meromorphically wherever the products are defined. This theorem is **proved here** and **new relative to the pinned snapshot**.*

Proof. Assume first $|q| < 1$. By (2.8),

$$\begin{aligned} M_{1/q}(-t)^{-1} &= \prod_{j=1}^{\infty} E((1 - q^{-1})q^j(-t)) \\ &= \prod_{j=1}^{\infty} E((1 - q)q^{j-1}t) = M_q(t). \end{aligned}$$

The case $|q| > 1$ follows by interchanging q and $1/q$. Combining the first identity with reflection (2.11) gives the centered identity. $\square$

At the coefficient level, (4.6) says

$$\sum_{k=0}^n \binom{n}{k} m_k(q) (-1)^{n-k} m_{n-k}(1/q) = \delta_{n0}. \quad (4.7)$$

Thus the reciprocal moment row is the signed binomial-convolution inverse of the original row.

4.3 Cayley coordinate and the group action

Introduce the Cayley coordinate

$$c(q) = \frac{1+q}{1-q}. \quad (4.8)$$

A direct calculation gives

$$c(-q) = c(q)^{-1}, \quad c(1/q) = -c(q). \quad (4.9)$$

The formal continuation of theorem 4.1 can be written entirely in centered form:

$$\boxed{C_{-q}(t) = C_q(c(q)t).} \quad (4.10)$$

For $q > 1$, $c(q) < 0$, but C_q is even, so only $|c(q)|$ matters. Together, (4.6) and (4.10) realize the Klein four-group by argument dilation and multiplicative inversion.

The two requested orbits therefore reduce to the following explicit formulas:

$$M_{-1/2}(t) = e^{-t} M_{1/2}(3t), \quad M_2(t) = \frac{e^t}{M_{1/2}(t)}, \quad M_{-2}(t) = \frac{e^{2t}}{M_{1/2}(3t)}, \quad (4.11)$$

$$M_{-1/4}(t) = e^{-t/3} M_{1/4}(5t/3), \quad M_4(t) = \frac{e^t}{M_{1/4}(t)}, \quad M_{-4}(t) = \frac{e^{4t/3}}{M_{1/4}(5t/3)}. \quad (4.12)$$

Every coefficient identity among the eight requested cases follows from these six transformations.

First synthesis principle. The signs $q \mapsto -q$ and reciprocal scales $q \mapsto 1/q$ do not create eight unrelated special functions. They create two four-point parameter orbits. The positive contraction determines the negative probability law by affine transport and determines both reciprocal germs by convolution inversion.

5 Spectral q^2 -Pochhammer products and generalized Rvachev laws

5.1 The hyperbolic product of the moment germ

Euler's product for $\sinh$ gives

$$S(z) = \frac{\sinh(z/2)}{z/2} = \prod_{k=1}^{\infty} \left(1 + \frac{z^2}{4\pi^2 k^2}\right). \quad (5.1)$$

Because $\sum_{j \geq 0} (1-q)q^j = 1$, the exponential pieces of (2.6) combine to $e^{t/2}$. Rearranging the remaining normally convergent product yields the spectral factorization.

Theorem 5.1 (Spectral Pochhammer factorization). *For $|q| < 1$,*

$$M_q(t) = e^{t/2} \prod_{k=1}^{\infty} \left(-\frac{(1-q)^2 t^2}{4\pi^2 k^2}; q^2 \right)_{\infty}. \quad (5.2)$$

For $|q| > 1$,

$$M_q(t) = e^{t/2} \prod_{k=1}^{\infty} \left(-\frac{(1-q^{-1})^2 t^2}{4\pi^2 k^2}; q^{-2} \right)_{\infty}^{-1}. \quad (5.3)$$

The theorem is **proved here**.

Proof. Insert (5.1) into $E((1-q)q^j t) = e^{(1-q)q^j t/2} S((1-q)q^j t)$. For fixed k , the product over j is

$$\prod_{j \geq 0} \left(1 + \frac{(1-q)^2 q^{2j} t^2}{4\pi^2 k^2}\right) = \left(-\frac{(1-q)^2 t^2}{4\pi^2 k^2}; q^2 \right)_{\infty}.$$

This proves (5.2). In the reciprocal product, set $r = q^{-1}$. The scale $(1-q)q^{-j} = -(1-r)r^{j-1}$; its square and the inverse product give (5.3). The exponential normalization is again $e^{t/2}$. $\square$

This theorem resolves an apparent ambiguity in the requested parameters. The geometric ratio is q , but the spectral Pochhammer nome is q^2 . Therefore

$$q = \pm \frac{1}{2} \implies \text{spectral nome } \frac{1}{4}, \quad q = \pm \frac{1}{4} \implies \text{spectral nome } \frac{1}{16}.$$

Their reciprocal partners ± 2 and ± 4 use the same contracting nomes but with inverse factors. Inversion is literally a zero-pole exchange.

5.2 The centered Rvachev random series

Let

$$R_q = \sum_{j=0}^{\infty} q^j V_j, \quad V_j = U_j - \frac{1}{2}, \quad |q| < 1. \quad (5.4)$$

Then

$$Y_q = \frac{1}{2} + (1-q)R_q. \quad (5.5)$$

Under the Fourier convention $\hat{\mu}(z) = \int e^{2\pi i z x} d\mu(x)$, its characteristic function is

$$\Phi_q(z) = \mathbb{E} e^{2\pi i z R_q} = \prod_{j=0}^{\infty} \text{sinc}(q^j z), \quad \text{sinc } z = \frac{\sin \pi z}{\pi z}. \quad (5.6)$$

At $q = 1/2$, R_q is the random variable whose density is Rvachev's up-function in the standard support normalization. Since the V_j are symmetric,

$$R_{-a} \stackrel{d}{=} R_a, \quad 0 < a < 1. \quad (5.7)$$

Thus the sign dependence of Y_q is entirely caused by the outer affine normalization $1 - q$.

The Fourier product itself has the Pochhammer form

$$\Phi_q(z) = \prod_{k=1}^{\infty} (z^2/k^2; q^2)_{\infty}. \quad (5.8)$$

Its Mahler equation is

$$\Phi_q(z) = \text{sinc}(z) \Phi_q(qz). \quad (5.9)$$

5.3 Integer reciprocal scales and zero multiplicities

The classical dyadic transform has zero multiplicity $1 + \nu_2(n)$ at every nonzero integer n . The same argument works for every integer reciprocal scale.

Proposition 5.2 (Base- b zero divisor). *Let $q = 1/b$ with integer $b \geq 2$. At a nonzero integer n ,*

$$\text{ord}_{z=n} \Phi_{1/b}(z) = 1 + \nu_b(n), \quad (5.10)$$

where $\nu_b(n) = \max\{j : b^j \mid n\}$. This is [proved here](#).

Proof. The factor $\text{sinc}(z/b^j)$ has a simple zero at $z = n$ exactly when n/b^j is a nonzero integer, that is, when $0 \leq j \leq \nu_b(n)$. All other factors are nonzero there. $\square$

For $b = 4$,

$$\text{ord}_{z=n} \Phi_{1/4}(z) = 1 + \left\lfloor \frac{\nu_2(n)}{2} \right\rfloor. \quad (5.11)$$

The quartic transform therefore sees pairs of binary valuation levels at once. This is the spectral counterpart of the even/odd scale decimation in the next section.

6 Scale multisection and q -integer convolution

6.1 Decomposing scales modulo m

Define the q -integer

$$[m]_q = 1 + q + \cdots + q^{m-1} = \frac{1 - q^m}{1 - q}. \quad (6.1)$$

Separate the coordinate index in (2.1) as $j = mk + r$.

Theorem 6.1 (Geometric multisection). *Let $m \geq 2$, $|q| \neq 1$, and $[m]_q \neq 0$. As analytic germs,*

$$M_q(t) = \prod_{r=0}^{m-1} M_{q^m} \left(\frac{q^r}{[m]_q} t \right). \quad (6.2)$$

For $|q| < 1$, this is the distributional identity

$$Y_q \stackrel{d}{=} \sum_{r=0}^{m-1} \frac{q^r}{[m]_q} Y_{q^m}^{(r)}, \quad (6.3)$$

where the variables on the right are independent. The theorem is [proved here](#) and [new relative to the pinned snapshot](#) in this normalized form.

Proof. For $|q| < 1$,

$$\begin{aligned} Y_q &= (1-q) \sum_{r=0}^{m-1} \sum_{k \geq 0} q^{mk+r} U_{mk+r} \\ &= \sum_{r=0}^{m-1} \frac{(1-q)q^r}{1-q^m} \left((1-q^m) \sum_{k \geq 0} (q^m)^k U_{mk+r} \right), \end{aligned}$$

which is (6.3). Independence gives (6.2). The two sides are normalized formal solutions and the identity extends to the reciprocal regime by the product definition, or directly by multisection of the contracting reciprocal product. $\square$

For $m = 2$,

$$Y_q \stackrel{d}{=} \frac{1}{1+q} Y_{q^2}^{(0)} + \frac{q}{1+q} Y_{q^2}^{(1)}. \quad (6.4)$$

The requested specializations include

$$Y_{1/2} \stackrel{d}{=} \frac{2}{3} Y_{1/4}^{(0)} + \frac{1}{3} Y_{1/4}^{(1)}, \quad (6.5)$$

$$Y_{-1/2} \stackrel{d}{=} 2Y_{1/4}^{(0)} - Y_{1/4}^{(1)}, \quad (6.6)$$

$$M_2(t) = M_4(t/3)M_4(2t/3). \quad (6.7)$$

The first identity decomposes the classical Fabius distribution into two independent quartic distributions at unequal scales. The second is a signed version of the same decomposition and agrees with $Y_{-1/2} \stackrel{d}{=} 3Y_{1/2} - 1$.

6.2 Rvachev multisection

The unnormalized centered series has the cleaner identity

$$\boxed{R_q \stackrel{d}{=} \sum_{r=0}^{m-1} q^r R_{q^m}^{(r)}} \quad \boxed{\Phi_q(z) = \prod_{r=0}^{m-1} \Phi_{q^m}(q^r z)}. \quad (6.8)$$

At $q = 1/2$, $m = 2$, the dyadic up-density is therefore a convolution of two quartic up-densities, one compressed by two. In density notation,

$$u_{1/2} = u_{1/4} * (2u_{1/4}(2 \cdot)). \quad (6.9)$$

At the product level, all these identities are instances of the elementary Pochhammer multisection

$$(a; q)_\infty = \prod_{r=0}^{m-1} (aq^r; q^m)_\infty. \quad (6.10)$$

Second synthesis principle. The classical dyadic object is not isolated from the quartic object. It is an exact two-component weighted independent-sum decomposition at the random-variable level, a convolution at the density level, a factorization at the Fourier level, and a multisection identity at the Pochhammer level. Negative and reciprocal parameters preserve the same diagram, with signed coefficients or inverse factors as appropriate.

7 Reciprocal Laplace duality and vertical moment geometry

7.1 From inverse hyperbolic products to Laplace laws

The negative formal variance for $q > 1$ suggests that the reciprocal germ should not be sought on the real line. The centered identity $C_q = 1/C_{1/q}$ turns that obstruction into a positive characteristic function.

Let $r = 1/q$, so $|r| < 1$, and define

$$b_{j,k}(r) = \frac{|1-r||r|^j}{2\pi k}, \quad j \geq 0, k \geq 1. \quad (7.1)$$

Let $Z_{j,k}$ be independent standard symmetric Laplace variables with density $\frac{1}{2}e^{-|x|}$ and characteristic function $(1+t^2)^{-1}$.

Theorem 7.1 (Positive Laplace representation of the reciprocal germ). *For every real q with $|q| > 1$, the series*

$$L_{1/q} = \sum_{j=0}^{\infty} \sum_{k=1}^{\infty} b_{j,k}(1/q) Z_{j,k} \quad (7.2)$$

converges in L^2 and almost surely, and

$$C_q(t) = \mathbb{E} e^{itL_{1/q}}. \quad (7.3)$$

Consequently

$$m_n(q) = \mathbb{E} \left(\frac{1}{2} + iL_{1/q} \right)^n. \quad (7.4)$$

This theorem is [proved here](#) and [new relative to the pinned snapshot](#).

Proof. By (5.2) and reciprocity,

$$\begin{aligned} C_q(t) &= \frac{1}{C_r(t)} \\ &= \prod_{j \geq 0} \prod_{k \geq 1} \left(1 + \frac{(1-r)^2 r^{2j} t^2}{4\pi^2 k^2} \right)^{-1} \\ &= \prod_{j,k} (1 + b_{j,k}(r)^2 t^2)^{-1}. \end{aligned}$$

Moreover,

$$\sum_{j,k} b_{j,k}(r)^2 = \frac{(1-r)^2}{4\pi^2(1-r^2)} \zeta(2) < \infty.$$

Since a standard Laplace variable has variance 2, Kolmogorov's convergence criterion gives L^2 and almost-sure convergence of (7.2). The characteristic functions of the finite partial sums converge to the displayed product, proving (7.3). Finally, $M_q(t) = e^{t/2} C_q(t) = \mathbb{E} \exp(t(1/2 + iL_r))$ near the origin, and coefficient comparison proves (7.4). $\square$

The theorem replaces the phrase “signed probability” by a precise geometric object: a positive measure on the vertical line

$$\left\{ \frac{1}{2} + ix : x \in \mathbb{R} \right\}. \quad (7.5)$$

The reciprocal moments are ordinary polynomial moments of that measure.

7.2 The zeta bridge in the Laplace cumulants

For a Laplace variable of scale b , the even cumulants are $(2m)!b^{2m}/m$. Hence

$$\kappa_{2m}(L_r) = \frac{(2m)!}{m} \frac{(1-r)^{2m}}{(2\pi)^{2m}(1-r^{2m})} \zeta(2m). \quad (7.6)$$

Using

$$B_{2m} = (-1)^{m+1} \frac{2(2m)!}{(2\pi)^{2m}} \zeta(2m) \quad (7.7)$$

and $q = 1/r$, one recovers

$$\kappa_{2m}(q) = (-1)^m \kappa_{2m}(L_r), \quad (7.8)$$

exactly as required by the factor i^{2m} in (7.4). This is a direct three-way bridge among Bernoulli numbers, the Riemann zeta values, and the reciprocal q -Fabius germ.

7.3 Exact Hankel signature

Let

$$\Delta_n(q) = \det (m_{i+j}(q))_{0 \leq i,j < n}. \quad (7.9)$$

Theorem 7.2 (Hankel signature). *For every real $|q| > 1$ and $n \geq 1$,*

$$\boxed{\operatorname{sgn} \Delta_n(q) = (-1)^{\binom{n}{2}}}. \quad (7.10)$$

In particular, no Hankel determinant vanishes. The theorem is [proved here](#) and [new relative to the pinned snapshot](#).

Proof. Let $H_n(L_r)$ be the positive Hankel determinant of the real moments of L_r . It is strictly positive because the Laplace convolution has infinite support, so the monomials $1, x, \dots, x^{n-1}$ are linearly independent in L^2 of its law. Translation of a moment sequence leaves its Hankel determinant unchanged: the change-of-basis matrix from x^k to $(x+c)^k$ is unit lower triangular. Scaling the variable by a multiplies an n th Hankel determinant by $a^{n(n-1)}$. Applying these facts to $1/2 + iL_r$ gives

$$\Delta_n(q) = i^{n(n-1)} H_n(L_r) = (-1)^{\binom{n}{2}} H_n(L_r).$$

□

The exact determinants through order nine for $q = \pm 2, \pm 4$ are included in `data/reciprocal_hankel_signatures.csv`. Their sizes vary enormously, but their signs follow (7.10) without exception.

7.4 Orthogonal polynomials on a vertical line

Let $Q_n^{(r)}$ be the monic orthogonal polynomial of degree n for the real law of L_r . Symmetry gives a recurrence

$$Q_{n+1}^{(r)}(x) = xQ_n^{(r)}(x) - \beta_n^{(r)}Q_{n-1}^{(r)}(x), \quad \beta_n^{(r)} > 0. \quad (7.11)$$

Define

$$P_n^{(q)}(z) = i^n Q_n^{(1/q)}\left(\frac{z - 1/2}{i}\right). \quad (7.12)$$

Then $P_n^{(q)}$ is monic and orthogonal for the reciprocal moment functional. Its recurrence is

$$P_{n+1}^{(q)}(z) = (z - 1/2)P_n^{(q)}(z) + \beta_n^{(1/q)}P_{n-1}^{(q)}(z). \quad (7.13)$$

In the standard monic form $P_{n+1} = (z - a_n)P_n - b_n P_{n-1}$, the subdiagonal coefficient is $b_n = -\beta_n < 0$. Every zero has the form

$$z = \frac{1}{2} + ix, \quad x \in \mathbb{R}. \quad (7.14)$$

This supplies a concrete spectral problem for the reciprocal germs: determine the recurrence coefficients $\beta_n^{(r)}$, their asymptotics, and their relation to the geometric double lattice $b_{j,k}(r)$.

Third synthesis principle. Reciprocal q does not destroy positivity; it rotates it. The centered reciprocal germ is positive definite on the Fourier axis, the raw moments live on $\Re z = 1/2$, the Hankel form has a deterministic alternating signature, and its orthogonal zeros align on the same vertical line.

8 q -Fabius–Bernoulli Appell polynomials

8.1 Inverse geometric averaging

For $|r| < 1$, define the translation-averaging operator on polynomials by

$$(\mathcal{A}_r f)(x) = \mathbb{E}f(x + Y_r). \quad (8.1)$$

On formal exponentials, $\mathcal{A}_r$ is the multiplier $M_r(D)$:

$$\mathcal{A}_r e^{xt} = M_r(t) e^{xt}.$$

Reciprocity therefore identifies its inverse without solving a triangular system coefficient by coefficient:

$$\boxed{\mathcal{A}_r^{-1} = M_{1/r}(-D)}. \quad (8.2)$$

The reciprocal q -Fabius germ is precisely the deconvolution symbol for geometric-uniform averaging.

Definition 8.1 (q -Fabius–Bernoulli polynomials). For $|r| < 1$, define the Appell sequence $\mathfrak{B}_n^{(r)}(x)$ by

$$\boxed{\sum_{n=0}^{\infty} \mathfrak{B}_n^{(r)}(x) \frac{t^n}{n!} = \frac{e^{xt}}{M_r(t)}}. \quad (8.3)$$

Theorem 8.2 (Exact deconvolution). *For every polynomial degree n ,*

$$\boxed{\mathbb{E} \mathfrak{B}_n^{(r)}(x + Y_r) = x^n}. \quad (8.4)$$

The sequence is Appell:

$$\frac{d}{dx} \mathfrak{B}_n^{(r)}(x) = n \mathfrak{B}_{n-1}^{(r)}(x). \quad (8.5)$$

These statements are [proved here](#) and [new relative to the pinned snapshot](#).

Proof. Taking the expectation of the generating function after replacing x by $x + Y_r$ multiplies its right-hand side by $M_r(t)$, leaving e^{xt} . Coefficient comparison proves (8.4). Differentiating (8.3) with respect to x proves (8.5). $\square$

8.2 Bernoulli and concentration limits

At $r = 0$, $Y_0 = U_0$ and $M_0(t) = E(t)$. Therefore

$$\frac{e^{xt}}{M_0(t)} = \frac{te^{xt}}{e^t - 1}, \quad \boxed{\mathfrak{B}_n^{(0)}(x) = B_n(x)}, \quad (8.6)$$

where $B_n(x)$ is the ordinary Bernoulli polynomial with $B_1(x) = x - 1/2$. At the opposite probability boundary, $\text{Var}(Y_r) \rightarrow 0$ as $r \rightarrow 1^-$ and $M_r(t) \rightarrow e^{t/2}$ locally uniformly. Hence

$$\boxed{\mathfrak{B}_n^{(r)}(x) \longrightarrow (x - 1/2)^n \quad (r \rightarrow 1^-)}. \quad (8.7)$$

The family continuously interpolates Bernoulli deconvolution of one uniform average and trivial deconvolution of a point mass at $1/2$.

Reciprocity also explains the boundary from outside the unit disk. As $|q| \rightarrow \infty$, $1/q \rightarrow 0$ and

$$M_q(t) = \frac{e^t}{M_{1/q}(t)} \longrightarrow \frac{te^t}{e^t - 1} = \sum_{n \geq 0} B_n(1) \frac{t^n}{n!}. \quad (8.8)$$

Thus Bernoulli polynomials sit at both ends of the sign/reciprocal diagram: as the inverse of ordinary uniform averaging and as the limiting reciprocal moment germ.

8.3 A Bernoulli–Mahler refinement equation

For $0 < |r| < 1$, use the Mahler equation $M_r(t) = E((1-r)t)M_r(rt)$ in (8.3):

$$\begin{aligned} \frac{e^{xt}}{M_r(t)} &= \frac{(1-r)t}{e^{(1-r)t} - 1} \frac{e^{xt}}{M_r(rt)} \\ &= \left(\sum_{k \geq 0} B_k \frac{((1-r)t)^k}{k!} \right) \left(\sum_{j \geq 0} \mathfrak{B}_j^{(r)}(x/r) \frac{(rt)^j}{j!} \right). \end{aligned} \quad (8.9)$$

Here $B_1 = -1/2$ is the standard Bernoulli-number convention. Therefore

$$\boxed{\mathfrak{B}_n^{(r)}(x) = \sum_{k=0}^n \binom{n}{k} B_k (1-r)^k r^{n-k} \mathfrak{B}_{n-k}^{(r)}(x/r)}. \quad (8.10)$$

This relation is a Bernoulli-polynomial analogue of the Fabius Mahler equation. It combines a local uniform deconvolution at scale $1-r$ with the same Appell family at the contracted argument x/r .

8.4 Bell-polynomial form and first examples

Taking logarithms in (8.3) gives

$$\log \frac{e^{xt}}{M_r(t)} = (x - 1/2)t - \sum_{m \geq 1} \kappa_{2m}(r) \frac{t^{2m}}{(2m)!}. \quad (8.11)$$

Consequently

$$\boxed{\mathfrak{B}_n^{(r)}(x) = \mathcal{B}_n \left(x - \frac{1}{2}, -\kappa_2(r), 0, -\kappa_4(r), 0, \dots \right)}. \quad (8.12)$$

Writing $z = x - 1/2$ and $v_r = \kappa_2(r) = (1-r)/(12(1+r))$, one obtains

$$\begin{aligned} \mathfrak{B}_0^{(r)}(x) &= 1, & \mathfrak{B}_1^{(r)}(x) &= z, \\ \mathfrak{B}_2^{(r)}(x) &= z^2 - v_r, & \mathfrak{B}_3^{(r)}(x) &= z^3 - 3v_r z, \\ \mathfrak{B}_4^{(r)}(x) &= z^4 - 6v_r z^2 + 3v_r^2 - \kappa_4(r). \end{aligned} \quad (8.13)$$

The symmetry

$$\mathfrak{B}_n^{(r)}(1-x) = (-1)^n \mathfrak{B}_n^{(r)}(x) \quad (8.14)$$

follows immediately from centered evenness. Their arithmetic, root geometry, and possible analogues of the von Staudt–Clausen and multiplication theorems form a natural new research program.

9 A Fabius–Pochhammer moment polynomial

9.1 Definition and polynomial recurrence

The rational moments $a_n(q) = m_n(q)/n!$ have poles at roots of unity because of the factors $1 - q^j$ in the recurrence. Multiplying by the finite q -Pochhammer symbol and removing the obvious $(1 - q)^n$ scale produces a polynomial.

Definition 9.1. For $n \geq 0$, define

$$\mathcal{P}_n(q) = \frac{(q; q)_n}{(1 - q)^n} \frac{m_n(q)}{n!}. \quad (9.1)$$

At $q = 1$ the right-hand expression is interpreted by polynomial continuation.

Theorem 9.2 (Polynomial recurrence). *The functions $\mathcal{P}_n$ are polynomials in $\mathbb{Q}[q]$, with $\deg \mathcal{P}_n \leq \binom{n}{2}$, and*

$$\mathcal{P}_0(q) = 1, \quad (9.2)$$

$$\mathcal{P}_n(q) = \sum_{k=0}^{n-1} \frac{q^k}{(n-k+1)!} \frac{(q; q)_{n-1}}{(q; q)_k} \mathcal{P}_k(q). \quad (9.3)$$

Moreover,

$$\mathcal{P}_n(0) = \frac{1}{(n+1)!}, \quad [q^{\binom{n}{2}}] \mathcal{P}_n(q) = \frac{B_n(1)}{n!}. \quad (9.4)$$

The theorem is **proved here** and **new relative to the pinned snapshot**.

Proof. Multiply (2.10) by $(q; q)_{n-1}/(1 - q)^n$ and substitute $a_k = (1 - q)^k \mathcal{P}_k/(q; q)_k$. This gives (9.3). The quotient $(q; q)_{n-1}/(q; q)_k = \prod_{j=k+1}^{n-1} (1 - q^j)$ is polynomial. If $\deg \mathcal{P}_k \leq \binom{k}{2}$, the k th summand has degree at most

$$k + \sum_{j=k+1}^{n-1} j + \binom{k}{2} = \binom{n}{2},$$

so induction proves polynomiality and the degree bound.

At $q = 0$, only the $k = 0$ summand survives, giving $\mathcal{P}_n(0) = 1/(n+1)!$. For the leading coefficient, let $q \rightarrow \infty$. By (8.8), $m_n(q)/n! \rightarrow B_n(1)/n!$, while

$$\frac{(q; q)_n}{(1 - q)^n} \sim q^{\binom{n}{2}}.$$

The coefficient assertion follows. □

The first polynomials are

$$\begin{aligned} \mathcal{P}_0(q) &= 1, & \mathcal{P}_1(q) &= \frac{1}{2}, \\ \mathcal{P}_2(q) &= \frac{q+2}{12}, & \mathcal{P}_3(q) &= \frac{q^2+q+1}{24}, \end{aligned} \quad (9.5)$$

$$\mathcal{P}_4(q) = -\frac{(q^2+q+1)(q^4-2q^3-5q^2-3q-6)}{720}. \quad (9.6)$$

The vanishing of $B_n(1)$ for odd $n > 1$ explains the strict degree drop $\deg \mathcal{P}_n < \binom{n}{2}$ in odd degrees.

9.2 A cyclotomic divisibility pattern

The exact computations through degree 16 reveal a stronger pattern:

$$\prod_{\substack{3 \leq d \leq n \\ d \text{ odd}}} [d]_q \text{ appears to divide } \mathcal{P}_n(q), \quad [d]_q = 1 + q + \cdots + q^{d-1}. \quad (9.7)$$

It is visible already in (9.5)–(9.6), and continues in the computed factorizations of $\mathcal{P}_5, \dots, \mathcal{P}_{16}$. The exact polynomial, proposed divisor, and quotient data are recorded in `data/moment_polynomial_divisibility.csv`. The odd-cumulant vanishing suggests the mechanism, but common cyclotomic factors between $1 - q^{2^m}$ and $1 - q^d$ make a direct denominator-counting argument insufficient.

Conjecture 9.3 (Odd q -integer divisor). *For every $n \geq 3$,*

$$\prod_{\substack{3 \leq d \leq n \\ d \text{ odd}}} [d]_q \mid \mathcal{P}_n(q) \quad \text{in } \mathbb{Q}[q]. \quad (9.8)$$

*This is **conjectural/open**. It has been checked symbolically for $n \leq 16$.*

A proof may be accessible through a root-of-unity analysis of the centered Bell expansion, or through a cyclic version of the two-nome partition function in [section 10](#). The multiplicities of repeated cyclotomic factors in the product make the conjecture substantially stronger than separate vanishing at primitive odd roots.

9.3 Endpoint interpretation

The main reason for introducing $\mathcal{P}_n$ is that the exact endpoint formula of [section 14](#) becomes

$$G_q(q^n) = \frac{q^{\binom{n+1}{2}}}{(q; q)_n} \mathcal{P}_n(q) \quad (0 < q \leq 1/2). \quad (9.9)$$

Thus the finite q -Pochhammer denominator at the inverse-geometric endpoint is not an imported notation: it is the exact pole-clearing factor of the moment recurrence. At $q = 1/2$ it becomes the familiar product of Mersenne factors; at $q = 1/4$ it becomes a product of $4^j - 1$ factors.

10 A two-nome Pochhammer–Prouhet partition function

10.1 Definition on the binary cube

For a binary vector $\epsilon = (\epsilon_0, \dots, \epsilon_{N-1}) \in \{0, 1\}^N$, define its Hamming weight, position energy, and geometric projection by

$$|\epsilon| = \sum_{j=0}^{N-1} \epsilon_j, \quad \Omega(\epsilon) = \sum_{j=0}^{N-1} j\epsilon_j, \quad A_{\sigma, N}(\epsilon) = \sum_{j=0}^{N-1} \epsilon_j \sigma^j. \quad (10.1)$$

The usual Thue–Morse sign on the cube is $(-1)^{|\epsilon|}$.

Definition 10.1 (Two-nome binary-cube partition function). For parameters u, t, ρ, σ , set

$$\mathcal{Z}_N(u, t; \rho, \sigma) = \prod_{j=0}^{N-1} \left(1 - u \rho^j e^{t\sigma^j} \right). \quad (10.2)$$

Expanding one factor at a time gives the exact cube sum

$$\mathcal{Z}_N(u, t; \rho, \sigma) = \sum_{\epsilon \in \{0,1\}^N} (-u)^{|\epsilon|} \rho^{\Omega(\epsilon)} e^{tA_{\sigma,N}(\epsilon)}. \quad (10.3)$$

The four variables have distinct jobs: u marks Hamming weight, ρ marks position energy, t marks projected moments, and σ chooses the geometric projection. This separation is the cleanest way to connect Gaussian coefficients to Thue–Morse Prouhet sums without identifying their nomes by fiat.

10.2 Two boundary faces

At $t = 0$,

$$\mathcal{Z}_N(u, 0; \rho, \sigma) = (u; \rho)_N. \quad (10.4)$$

The finite q -binomial theorem then gives

$$(u; \rho)_N = \sum_{k=0}^N (-u)^k \rho^{\binom{k}{2}} \left[\begin{matrix} N \\ k \end{matrix} \right]_{\rho}. \quad (10.5)$$

At the opposite face $(u, \rho) = (1, 1)$,

$$\mathcal{Z}_N(1, t; 1, \sigma) = \prod_{j=0}^{N-1} (1 - e^{t\sigma^j}) = \sum_{\epsilon} (-1)^{|\epsilon|} e^{tA_{\sigma,N}(\epsilon)}. \quad (10.6)$$

This is the exponential generating function of the geometric Prouhet moments. Thus the q -Pochhammer and Thue–Morse products are not merely analogous: they are perpendicular faces of one four-parameter finite product.

Master binary-cube object. The finite Gaussian coefficient records how many selected digit positions carry a given total energy. The Prouhet generator records how the same subset projects to a geometric number. The partition function $\mathcal{Z}_N$ retains both statistics simultaneously and permits independent nomes ρ and σ .

10.3 Simultaneous multisection

Splitting the index $j = mk + r$ gives a multisection in both nomes at once.

Theorem 10.2 (Two-nome multisection). *For integers $m, N \geq 1$,*

$$\mathcal{Z}_{mN}(u, t; \rho, \sigma) = \prod_{r=0}^{m-1} \mathcal{Z}_N(u\rho^r, t\sigma^r; \rho^m, \sigma^m). \quad (10.7)$$

The identity is *proved here* and *new relative to the pinned snapshot*.

Proof. The factor with index $j = mk + r$ on the left is

$$1 - u\rho^{mk+r} e^{t\sigma^{mk+r}} = 1 - (u\rho^r)(\rho^m)^k e^{(t\sigma^r)(\sigma^m)^k},$$

which is the k th factor of the r th product on the right. $\square$

At $t = 0$, (10.7) is finite Pochhammer multisection. At $(u, \rho) = (1, 1)$, it is the derivative-comb convolution law developed in [section 13](#). At $N \rightarrow \infty$ in the normalized Prouhet jet, it becomes the geometric-uniform multisection theorem [6.1](#).

10.4 Mixed Gaussian–Prouhet moments

Define

$$\mathcal{M}_{N,k,m}(\rho, \sigma) = \sum_{|\epsilon|=k} \rho^{\Omega(\epsilon)} A_{\sigma,N}(\epsilon)^m. \quad (10.8)$$

From (10.3),

$$\boxed{\mathcal{M}_{N,k,m}(\rho, \sigma) = (-1)^k m! [u^k t^m] \mathcal{Z}_N(u, t; \rho, \sigma).} \quad (10.9)$$

This coefficient simultaneously refines a Gaussian row by geometric power moments. For $m = 0$ it is the position-energy layer $\rho^{\binom{k}{2}} \left[\begin{smallmatrix} N \\ k \end{smallmatrix} \right]_{\rho}$. Summing over k with sign recovers the Prouhet moment in [section 12](#).

10.5 Root-of-unity cancellation beyond classical Prouhet

The order of $\mathcal{Z}_N$ at $t = 0$ can be read factor by factor.

Proposition 10.3 (Cancellation order). *Assume $\sigma \neq 0$. Then*

$$\text{ord}_{t=0} \mathcal{Z}_N(u, t; \rho, \sigma) = \# \{0 \leq j < N : u\rho^j = 1\}. \quad (10.10)$$

*Consequently the corresponding weighted mixed moments vanish below this order. The proposition is **proved here**.*

Proof. At $t = 0$, the j th factor is $1 - u\rho^j$. If it is nonzero, it contributes no zero. If $u\rho^j = 1$, the factor is $1 - e^{t\sigma^j} = -\sigma^j t + O(t^2)$ and has one simple zero. Orders add under multiplication. $\square$

Classical Prouhet cancellation is the maximal case $u = \rho = 1$, in which all N factors vanish. At $u = 1, \rho = -1$, only even positions contribute zeros, so the order is $\lceil N/2 \rceil$. More generally, root-of-unity choices of ρ select arithmetic progressions of digit positions. This gives a finite, exactly controlled family of “partial Prouhet” cancellations that is absent if the Gaussian and geometric nomes are collapsed too early.

10.6 The Fabius MGF as a renormalized Prouhet jet

Set $(u, \rho, \sigma) = (1, 1, q)$ and replace t by $(1 - q)t$. Since $1 - e^z = -zE(z)$,

$$\mathcal{Z}_N(1, (1 - q)t; 1, q) = (-1)^N (1 - q)^N q^{\binom{N}{2}} t^N \prod_{j=0}^{N-1} E((1 - q)q^j t). \quad (10.11)$$

Therefore, for $0 < |q| < 1$,

$$\boxed{M_q(t) = \lim_{N \rightarrow \infty} \frac{\mathcal{Z}_N(1, (1 - q)t; 1, q)}{(-1)^N (1 - q)^N q^{\binom{N}{2}} t^N}.} \quad (10.12)$$

The Fabius MGF is the normalized residual jet left after removing the maximal order- N Prouhet zero. This identity makes the transition from a finite Thue–Morse cancellation block to the smooth infinite Fabius distribution completely explicit.

11 Signed position-energy layers and negative Gaussian coefficients

11.1 The Gaussian layer formula

At fixed Hamming weight k , define

$$H_{N,k}(\rho) = \sum_{|\epsilon|=k} \rho^{\Omega(\epsilon)}. \quad (11.1)$$

Extracting u^k in (10.5) gives

$$H_{N,k}(\rho) = \rho^{\binom{k}{2}} \begin{bmatrix} N \\ k \end{bmatrix}_{\rho}. \quad (11.2)$$

This is the principal specialization of the Schur function of a rectangular partition, but its binary-cube meaning is more useful here: it is the exact partition function of k selected bit positions with linear energy j .

Complementing selected positions or reversing them gives the reciprocity

$$H_{N,k}(\rho) = \rho^{k(N-1)} H_{N,k}(\rho^{-1}). \quad (11.3)$$

The Gaussian coefficient itself satisfies

$$\begin{bmatrix} N \\ k \end{bmatrix}_{\rho} = \rho^{k(N-k)} \begin{bmatrix} N \\ k \end{bmatrix}_{\rho^{-1}}. \quad (11.4)$$

11.2 Exact sign laws at negative nome

For $0 < a < 1$, every factor $1 - (-a)^j$ is positive, so $\begin{bmatrix} N \\ k \end{bmatrix}_{-a} > 0$. Equation (11.2) gives

$$\operatorname{sgn} H_{N,k}(-a) = (-1)^{\binom{k}{2}} \quad (0 < a < 1). \quad (11.5)$$

For $a > 1$, use reciprocity and the positive coefficient at $-1/a$:

$$\operatorname{sgn} H_{N,k}(-a) = (-1)^{\binom{k}{2} + k(N-k)} \quad (a > 1). \quad (11.6)$$

Thus the four signs in each requested orbit are controlled by two elementary quadratic parities: the internal staircase $\binom{k}{2}$ and the rectangular complement $k(N-k)$.

11.3 Even/odd position splitting

Negative nome separates even and odd positions. Pairing the factors gives

$$(z; -a)_{2m} = (z; a^2)_m (-az; a^2)_m, \quad (11.7)$$

$$(z; -a)_{2m+1} = (z; a^2)_{m+1} (-az; a^2)_m. \quad (11.8)$$

This is the finite counterpart of the affine law for Y_{-a} and the decimation of Y_q into q^2 components. The negative Gaussian row is a convolution of two positive a^2 rows, one carrying an extra sign and one possibly an extra position.

11.4 The finite-field/unitary square

For a prime power Q , the positive Gaussian coefficient is the number of k -planes in $\mathbb{F}_Q^N$:

$$\begin{bmatrix} N \\ k \end{bmatrix}_Q = \# \operatorname{Gr}(k, \mathbb{F}_Q^N). \quad (11.9)$$

Let $d = k(N-k)$. Reciprocity gives

$$\begin{bmatrix} N \\ k \end{bmatrix}_{1/Q} = Q^{-d} \# \operatorname{Gr}(k, \mathbb{F}_Q^N). \quad (11.10)$$

Fu, Reiner, Stanton, and Thiem proved that negative Gaussian coefficients count nondegenerate subspaces of a Hermitian space. In their normalization,

$$\begin{bmatrix} N \\ k \end{bmatrix}_{-Q} = (-Q)^{-d} U_Q(N, k), \quad (11.11)$$

where $U_Q(N, k)$ is the number of nondegenerate k -subspaces of a nondegenerate Hermitian N -space over $\mathbb{F}_{Q^2}$. Combining this with reciprocity gives

$$\begin{bmatrix} N \\ k \end{bmatrix}_{-1/Q} = Q^{-2d} U_Q(N, k). \quad (11.12)$$

The requested parameters therefore form the square

nome	positive geometry	negative geometry
$\pm 2, \pm 1/2$	Grassmannians over $\mathbb{F}_2$	Hermitian geometry over $\mathbb{F}_4$
$\pm 4, \pm 1/4$	Grassmannians over $\mathbb{F}_4$	Hermitian geometry over $\mathbb{F}_{16}$

Multiplying by $\rho^{\binom{k}{2}}$ transports these counts to the position-energy layers $H_{N,k}(\rho)$. The file `data/grassmannian_unitary_square.csv` records exact examples through $N = 7$ and checks both reciprocal normalizations with integer arithmetic.

Example 11.1. For $Q = 2$, $N = 4$, $k = 2$, one has

$$\# \text{Gr}(2, \mathbb{F}_2^4) = 35, \quad U_2(4, 2) = 240.$$

Hence

$$\begin{bmatrix} 4 \\ 2 \end{bmatrix}_{1/2} = \frac{35}{16}, \quad \begin{bmatrix} 4 \\ 2 \end{bmatrix}_{-1/2} = \frac{15}{16}.$$

The second rational number is $240/2^8$; the extra power relative to the positive reciprocal reflects the unitary normalization Q^{-2d} .

12 A geometric q -Prouhet moment-transfer theorem

12.1 Signed moments of geometric subset sums

Define

$$S_{N,m}(q) = \sum_{\epsilon \in \{0,1\}^N} (-1)^{|\epsilon|} A_{q,N}(\epsilon)^m. \quad (12.1)$$

At $q = 2$, these are the usual Thue–Morse signed power sums on the integer block $0, 1, \dots, 2^N - 1$ after binary decoding. At $q = 1/2$ they are signed moments of a finite dyadic subset grid. At $q = 1/4$ they live on a finite base-4 Cantor skeleton.

The generating product is the Prouhet face (10.6):

$$\sum_{m \geq 0} S_{N,m}(q) \frac{t^m}{m!} = \prod_{j=0}^{N-1} (1 - e^{q^j t}). \quad (12.2)$$

Let

$$T_{q,N} = \sum_{j=0}^{N-1} q^j U_j. \quad (12.3)$$

Its MGF is $\prod_{j < N} E(q^j t)$.

Theorem 12.1 (Exact q -Prouhet moment transfer). *For every nonzero complex q , every $N \geq 1$, and every $\ell \geq 0$,*

$$S_{N,m}(q) = 0 \quad (0 \leq m < N), \quad (12.4)$$

$$S_{N,N+\ell}(q) = \boxed{(-1)^N \frac{(N+\ell)!}{\ell!} q^{\binom{N}{2}} \mathbb{E} T_{q,N}^\ell}. \quad (12.5)$$

The theorem is **proved here** and **new relative to the pinned snapshot**.

Proof. Factor each term on the right of (12.2) as $1 - e^{q^j t} = -q^j t E(q^j t)$. Then

$$\prod_{j=0}^{N-1} (1 - e^{q^j t}) = (-1)^N q^{\binom{N}{2}} t^N \prod_{j=0}^{N-1} E(q^j t). \quad (12.6)$$

The explicit factor t^N proves the vanishings. The coefficient of $t^{N+\ell}$ on the right is $(-1)^N q^{\binom{N}{2}} \mathbb{E} T_{q,N}^\ell / \ell!$, whereas on the left it is $S_{N,N+\ell} / (N+\ell)!$. $\square$

The first surviving moment is

$$S_{N,N}(q) = (-1)^N N! q^{\binom{N}{2}}. \quad (12.7)$$

The theorem not only states an order of cancellation; it identifies every post-cancellation coefficient with a positive-uniform moment when $q > 0$.

12.2 Recovering infinite q -Fabius moments

For $0 < |q| < 1$, $T_{q,N}$ converges in every L^p to $T_q = Y_q / (1 - q)$. Rearranging (12.5) yields

$$m_\ell(q) = (1 - q)^\ell \lim_{N \rightarrow \infty} (-1)^N q^{-\binom{N}{2}} \frac{\ell!}{(N+\ell)!} S_{N,N+\ell}(q). \quad (12.8)$$

Thus every Fabius or quartic moment is a normalized limit of the first nonvanishing jets of finite Thue–Morse blocks. This formula is the moment version of the full MGF limit (10.12).

The Python verification evaluates both sides of (12.5) exactly for all eight requested q values, $1 \leq N \leq 7$, and $0 \leq \ell \leq 4$. All residuals in `data/q_prouhet_verification.csv` are identically zero.

12.3 Sign and reciprocal transport of the finite comb

Introduce the signed atomic measure

$$\Delta_{q,N} = \sum_{\epsilon} (-1)^{|\epsilon|} \delta_{A_{q,N}(\epsilon)}. \quad (12.9)$$

Its bilateral Laplace transform is (12.2). Two elementary digit bijections describe the full sign/reciprocal orbit.

For $q = -a$, $0 < a < 1$, complement the odd-position digits. If $m = \lfloor N/2 \rfloor$ and

$$c_{a,N} = a \frac{1 - a^{2m}}{1 - a^2}, \quad (12.10)$$

then

$$\Delta_{-a,N} = (-1)^m \tau_{-c_{a,N}} \Delta_{a,N}, \quad (12.11)$$

where τ_c translates a measure by c . Digit reversal gives

$$\Delta_{q^{-1},N} = (D_{q^{-(N-1)}})_* \Delta_{q,N}, \quad (12.12)$$

where $D_c(x) = cx$. Hence the finite Thue–Morse comb itself carries the same Klein-four geometry as the infinite moment germ: sign is translation (up to a global parity), inversion is dilation.

13 Box-spline derivative combs and the quartic Cantor skeleton

13.1 Finite centered uniform sums

For $0 < q < 1$ and $N \geq 1$, let

$$R_{q,N} = \sum_{j=0}^{N-1} q^j V_j, \quad A_{q,N}^{\text{rad}} = \frac{1}{2} \sum_{j=0}^{N-1} q^j. \quad (13.1)$$

The law of $R_{q,N}$ is a nonuniform univariate box spline. Repeated convolution of interval indicators gives the exact truncated-power formula.

Proposition 13.1 (Finite box-spline formula). *The density $B_{q,N}$ of $R_{q,N}$ is*

$$B_{q,N}(x) = \frac{1}{(N-1)! q^{\binom{N}{2}}} \sum_{\epsilon \in \{0,1\}^N} (-1)^{|\epsilon|} \left(x + A_{q,N}^{\text{rad}} - A_{q,N}(\epsilon) \right)_+^{N-1}. \quad (13.2)$$

Distributionally,

$$D^N B_{q,N} = q^{-\binom{N}{2}} \sum_{\epsilon} (-1)^{|\epsilon|} \delta_{-A_{q,N}^{\text{rad}} + A_{q,N}(\epsilon)}. \quad (13.3)$$

The proposition is [proved here](#) and [derived from corpus infrastructure](#).

Proof. The density of $q^j V_j$ is q^{-j} times the indicator of $[-q^j/2, q^j/2]$. The standard inclusion–exclusion formula for a convolution of N interval densities with widths q^j gives (13.2); the product of widths is $q^{\binom{N}{2}}$. Since $D^N((x-a)_+^{N-1}/(N-1)!) = \delta_a$, differentiating termwise proves (13.3). $\square$

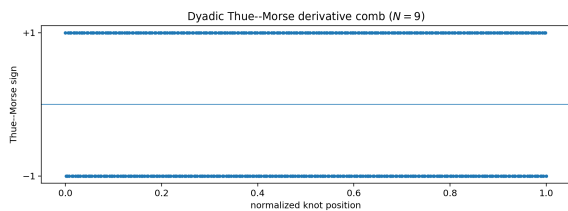
The derivative comb is exactly the centered version of $\Delta_{q,N}$. Therefore the q -Prouhet cancellation theorem is also the statement that its first $N-1$ distributional moments vanish. Repeated integration converts this signed atomic cancellation into a positive spline density.

13.2 Dyadic fullness versus quartic sparsity

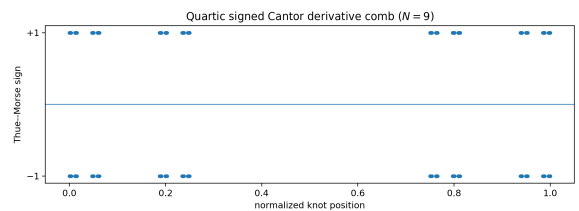
At $q = 1/2$, the subset sums $A_{1/2,N}(\epsilon)$ form a complete dyadic grid after a common scaling, and the signs are the Thue–Morse sequence in binary order. At $q = 1/4$, the digits occupy only the symbols 0 and 1 in base four. In the infinite limit the knot closure is the self-similar set

$$K_4^{\text{raw}} = \left\{ \sum_{j \geq 0} \epsilon_j 4^{-j} : \epsilon_j \in \{0,1\} \right\}, \quad \dim_{\text{H}} K_4^{\text{raw}} = \frac{\log 2}{\log 4} = \frac{1}{2}. \quad (13.4)$$

The finite derivative comb is therefore a signed Cantor skeleton, even though the infinite convolution density itself is smooth and has full interval support.



(a) $q = 1/2$: full dyadic grid.



(b) $q = 1/4$: base-4 digits 0, 1.

Figure 2: Signed derivative combs at level $N = 9$. The ordinate is the Thue–Morse sign. The quartic knots already display the dimension-1/2 Cantor geometry.

This contrast is conceptually important. The derivative skeleton remembers the discrete digit set, but integration through arbitrarily many uniform scales erases the gaps. Smoothness is not in conflict with a fractal hierarchy of singular finite derivatives.

13.3 Comb multisection

Equation (10.7) at $(u, \rho) = (1, 1)$ implies the convolution law

$$\Delta_{q,mN} = \ast_{r=0}^{m-1} (D_{q^r}) \ast \Delta_{q^m,N}. \quad (13.5)$$

For $m = 2$,

$$\Delta_{q,2N} = \Delta_{q^2,N} \ast (D_q) \ast \Delta_{q^2,N}. \quad (13.6)$$

At $q = 1/2$, the full dyadic Thue–Morse comb factors as a convolution of two quartic Cantor combs. This finite singular identity is the derivative-level ancestor of the smooth distributional convolution (6.5).

14 Exact inverse-geometric endpoint lattices

14.1 A simplex-volume identity

Let $0 < q < 1$, let G_q be the CDF of Y_q , and split the random series after n coordinates:

$$Y_q = (1 - q) \sum_{j=0}^{n-1} q^j U_j + q^n Y'_q, \quad (14.1)$$

where Y'_q is an independent copy of Y_q . At arguments of the form $q^n c$, the first n coordinates are constrained to a simplex. If the threshold is below the smallest box width, none of the upper constraints $U_j \leq 1$ is active.

Theorem 14.1 (Stop-loss endpoint formula). *Let $n \geq 1$ and*

$$0 \leq c \leq \frac{1 - q}{q}. \quad (14.2)$$

Then

$$G_q(q^n c) = \frac{q^{n(n+1)/2}}{(1 - q)^n n!} \mathbb{E} (c - Y_q)_+^n. \quad (14.3)$$

The theorem is [proved here](#) and [new relative to the pinned snapshot](#).

Proof. Condition on $Y'_q = y$ in (14.1). The event becomes

$$\sum_{j=0}^{n-1} (1 - q) q^j U_j \leq q^n (c - y).$$

If $c - y < 0$ its conditional probability is zero. Otherwise the condition (14.2) implies

$$q^n (c - y) \leq q^n c \leq (1 - q) q^{n-1},$$

which is no larger than the smallest coordinate width. Hence the intersection with the unit cube is the full coordinate simplex, of volume

$$\frac{q^{n^2} (c - y)^n}{n! \prod_{j=0}^{n-1} (1 - q) q^j} = \frac{q^{n(n+1)/2}}{(1 - q)^n n!} (c - y)^n.$$

Averaging over y proves the formula. $\square$

The right side is a normalized stop-loss moment. The parameter c therefore interpolates an entire small-endpoint family, not merely a single lattice. For $q > 1/2$, the admissible interval ends before $c = 1$; the general formula still holds, but the particularly simple moment specialization below does not.

14.2 The inverse-geometric moment identity

For $0 < q \leq 1/2$, $c = 1$ is admissible. Reflection of Y_q gives $1 - Y_q \stackrel{d}{=} Y_q$, so (14.3) becomes the promised generalization of the inverse-dyadic formula.

Corollary 14.2 (Inverse-geometric endpoint). *For $0 < q \leq 1/2$ and $n \geq 0$,*

$$G_q(q^n) = \frac{q^{\binom{n+1}{2}}}{(1-q)^n} \frac{m_n(q)}{n!}. \quad (14.4)$$

Equivalently,

$$G_q(q^n) = \frac{q^{\binom{n+1}{2}}}{(q; q)_n} \mathcal{P}_n(q). \quad (14.5)$$

This is **proved here** and **new relative to the pinned snapshot** for general q . The case $n = 0$ is immediate from $G_q(1) = 1$; the preceding theorem supplies every case $n \geq 1$.

At $q = 1/2$, the prefactor simplifies to $2^{-\binom{n}{2}}$ and one recovers the classical formula

$$F(2^{-n}) = 2^{-\binom{n}{2}} \frac{m_n(1/2)}{n!}. \quad (14.6)$$

At $q = 1/4$, one obtains a new exact inverse-quartic lattice:

$$G_{1/4}(4^{-n}) = \frac{4^{-\binom{n+1}{2}}}{(3/4)^n} \frac{m_n(1/4)}{n!}. \quad (14.7)$$

The first values are

$$G_{1/4}(1/4) = \frac{1}{6}, \quad G_{1/4}(1/16) = \frac{1}{240}, \quad G_{1/4}(1/64) = \frac{1}{51840}, \quad G_{1/4}(1/256) = \frac{121}{6768230400}. \quad (14.8)$$

The table `data/inverse_geometric_endpoint_values.csv` contains exact values through $n = 12$ for both $q = 1/2$ and $q = 1/4$.

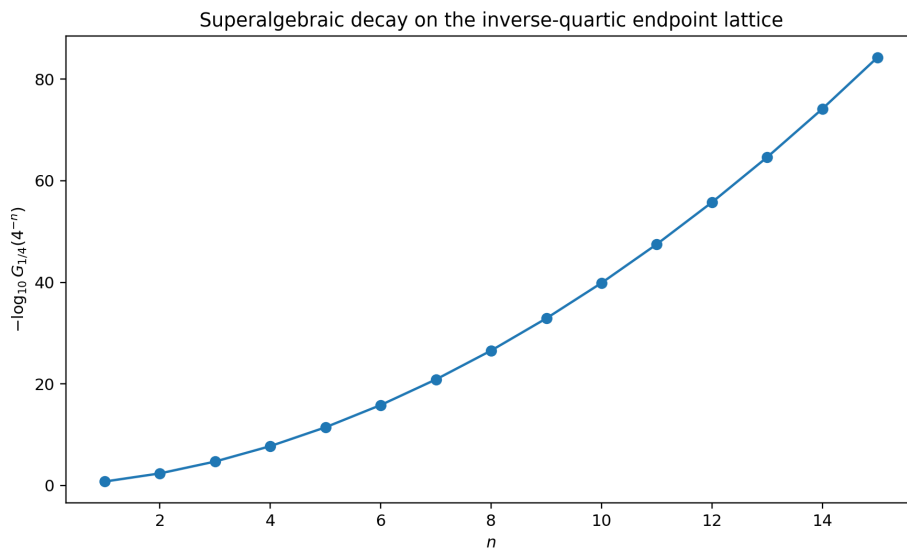


Figure 3: Exact inverse-quartic values plotted on a logarithmic ordinate. The quadratic growth of $-\log G_{1/4}(4^{-n})$ is the discrete shadow of the endpoint asymptotic in section 15.

14.3 All derivative jets on the lattice

Conditioning on the first coordinate gives the density equation

$$G'_q(x) = \frac{1}{1-q} \left[G_q\left(\frac{x}{q}\right) - G_q\left(\frac{x-1+q}{q}\right) \right]. \quad (14.9)$$

For $0 \leq x \leq 1-q$, the second term is zero. Iterating at $x = q^n$ yields an entire derivative jet.

Theorem 14.3 (Endpoint derivative jet). *Let $0 < q \leq 1/2$, $n \geq 0$, and $0 \leq r \leq n$. Then*

$$G_q^{(r)}(q^n) = (1-q)^{-r} q^{-\binom{r}{2}} G_q(q^{n-r}). \quad (14.10)$$

Consequently

$$G_q^{(r)}(q^n) = \frac{q^{\binom{n-r+1}{2} - \binom{r}{2}} m_{n-r}(q)}{(1-q)^n (n-r)!}. \quad (14.11)$$

The theorem is **proved here** and **new relative to the pinned snapshot**.

Proof. On the low interval, (14.9) reads $G'_q(x) = (1-q)^{-1} G_q(x/q)$. Differentiating $r-1$ more times and iterating produces a factor

$$(1-q)^{-r} q^{-(0+1+\dots+r-1)}.$$

At $x = q^n$, all intermediate points remain in the low interval because $q \leq 1/2$; at the boundary case $q = 1/2$, the differentiated endpoint identities follow by the flat matching of the two smooth branches. This proves (14.10); substitute (14.4) for the second formula. $\square$

For the density $g_q = G'_q$, the first case is

$$g_q(q^n) = \frac{q^{n(n-1)/2} m_{n-1}(q)}{(1-q)^n (n-1)!}. \quad (14.12)$$

At derivative order $r = n$, (14.10) reduces to the elementary value $(1-q)^{-n} q^{-\binom{n}{2}}$ because $G_q(1) = 1$.

14.4 Transport to negative parameters

Let G_{-a} denote the CDF of the negative law. From [theorem 4.1](#),

$$G_{-a}(x) = G_a\left(\frac{x + \beta_a}{\lambda_a}\right). \quad (14.13)$$

Therefore the positive endpoint lattice maps to

$$x_{a,n}^- = \lambda_a a^n - \beta_a, \quad \boxed{G_{-a}(x_{a,n}^-) = G_a(a^n)}. \quad (14.14)$$

For $a = 1/2$ these points are $3 \cdot 2^{-n} - 1$ and converge to the left endpoint -1 . For $a = 1/4$ they are $(5/3)4^{-n} - 1/3$ and converge to $-1/3$. All derivative jets transfer by the chain rule with factor λ_a^{-r} .

14.5 Successive endpoint ratios

Since the support of Y_q reaches 1, its moment ratios satisfy $m_{n+1}(q)/m_n(q) \rightarrow 1$. Dividing consecutive instances of (14.4) gives

$$\frac{G_q(q^{n+1})}{G_q(q^n)} = \frac{q^{n+1}}{(1-q)(n+1)} \frac{m_{n+1}(q)}{m_n(q)} \sim \frac{q^{n+1}}{(1-q)(n+1)}. \quad (14.15)$$

This ratio is much smaller than any fixed geometric rate; it is the first visible sign of the log-squared endpoint decay.

15 Rigorous endpoint and inverse asymptotics

15.1 A uniform two-term logarithmic law

The existing corpus contains a much sharper, all-orders, periodically corrected small- x expansion for the classical $q = 1/2$ function. Here we seek a result that is weaker but uniform in every fixed $0 < q \leq 1/2$, including $q = 1/4$. Put

$$L = \log(1/q), \quad \ell = \log(1/x). \quad (15.1)$$

Reflection and continuity imply $\mathbb{P}(Y_q \geq 1/2) = 1/2$. Hence

$$2^{-(n+1)} \leq m_n(q) \leq 1. \quad (15.2)$$

These elementary bounds are sufficient to determine the first two growing orders.

Theorem 15.1 (Uniform two-term endpoint asymptotic). *Fix $0 < q \leq 1/2$. As $x \downarrow 0$,*

$$\log G_q(x) = -\frac{\ell^2}{2L} - \frac{\ell \log \ell}{L} + \mathcal{O}_q(\ell). \quad (15.3)$$

*The theorem is **proved here** and **new relative to the pinned snapshot** for general q .*

Proof. First put $x = q^n$, so $\ell = nL$. Taking logarithms in (14.4) gives

$$\log G_q(q^n) = -\frac{L}{2}n(n+1) - n \log(1-q) - \log(n!) + \log m_n(q). \quad (15.4)$$

By (15.2), $\log m_n(q) = O(n)$. Stirling's formula gives $\log(n!) = n \log n - n + O(\log n)$. Substituting $n = \ell/L$ into (15.4) yields

$$\log G_q(q^n) = -\frac{\ell^2}{2L} - \frac{\ell \log \ell}{L} + O_q(\ell).$$

For arbitrary x , choose n such that $q^{n+1} < x \leq q^n$. Monotonicity places $G_q(x)$ between the two adjacent lattice values. Replacing ℓ by nL or $(n+1)L$ changes the two displayed leading terms by $O_q(\ell)$, so the same expansion holds throughout the interval. $\square$

The leading term alone gives the familiar log-normal-like flatness. The second term is larger than the unspecified linear correction and is therefore a genuinely stable feature of the whole geometric family.

For the two principal cases,

$$\log F(x) = -\frac{\log^2(1/x)}{2 \log 2} - \frac{\log(1/x) \log \log(1/x)}{\log 2} + O(\log(1/x)), \quad (15.5)$$

$$\log G_{1/4}(x) = -\frac{\log^2(1/x)}{4 \log 2} - \frac{\log(1/x) \log \log(1/x)}{2 \log 2} + O(\log(1/x)). \quad (15.6)$$

The quartic law is less flat at a common numerical x because its scale ladder descends more rapidly.

15.2 Asymptotic inversion

Let $y = G_q(x)$ and $S = \log(1/y)$. Equation (15.3) reads

$$S = \frac{\ell^2}{2L} + \frac{\ell \log \ell}{L} + O_q(\ell). \quad (15.7)$$

The first approximation is $\ell \sim \sqrt{2LS}$. One substitution then captures the logarithmic correction.

Corollary 15.2 (Inverse endpoint asymptotic). *For fixed $0 < q \leq 1/2$,*

$$\log \frac{1}{G_q^{-1}(y)} = \sqrt{2L \log(1/y)} - \frac{1}{2} \log \log(1/y) + O_q(1), \quad y \downarrow 0. \quad (15.8)$$

*This is **proved here**.*

Proof. Let $A = \sqrt{2LS}$. Set $\ell = A + \delta$. Inserting this into (15.7), the terms of size $A \log A$ are

$$\frac{A}{L}(\delta + \log A).$$

Thus $\delta = -\log A + O_q(1)$. Since $\log A = \frac{1}{2} \log S + O_q(1)$, the stated formula follows. Standard monotonicity of the left side of (15.7) controls the $O(1)$ remainder. $\square$

For negative $q = -a$, apply the affine inverse (14.13). Near the left endpoint $\ell_a = -a/(1-a)$,

$$G_{-a}^{-1}(y) - \ell_a = \lambda_a G_a^{-1}(y), \quad (15.9)$$

so the same square-root/log-log law holds after replacing distance to zero by distance to the left endpoint.

15.3 A Lambert- W_{-1} saddle coordinate

The all-orders classical analysis in the corpus uses the lower real branch of Lambert W . The same coordinate is predicted for every q , but proving the periodic correction and the complete saddle expansion uniformly in q is a separate project.

Let $\mathcal{L}_q(s) = M_q(-s) = \mathbb{E}e^{-sY_q}$ for $0 < q < 1$. From the product or from Mellin analysis of the scale sum, one expects the saddle equation

$$x \sim \frac{\log s}{s \log(1/q)}. \quad (15.10)$$

Solving its dominant part gives

$$s \sim -\frac{1}{x \log(1/q)} W_{-1}(-C_q x), \quad (15.11)$$

where $C_q > 0$ depends on the normalization chosen for the lower-order terms. The log-periodic fluctuation should have period one in $\log_{1/q} s$.

Conjecture 15.3 (General geometric all-orders endpoint expansion). *For each $0 < q < 1$, the density $g_q(x)$ admits as $x \downarrow 0$ a complete saddle expansion in inverse powers of $-W_{-1}(-C_q x)$, with analytic coefficient functions periodic in $\log_{1/q}(-W_{-1}(-C_q x)/x)$. The leading two unbounded terms agree with theorem 15.1. The conjecture is **conjectural/open** for general q ; the classical $q = 1/2$ case has substantially stronger support in the existing corpus.*

The quartic case is especially attractive. Scale decimation (6.5) couples its periodic correction to the classical one, but at two interlaced scales. This may allow the quartic periodic function to be reconstructed from the dyadic function through a nonlinear convolution saddle equation rather than recomputed from scratch.

16 Numerical experiments and reproducibility

16.1 Scope of the computation

The accompanying program `numerical_experiments.py` is intentionally self-contained. Exact calculations use Python's `Fraction` and integer arithmetic. The only floating-point components are independent high-precision product checks and deterministic Fourier inversions for the figures. No random sampling is used.

The script verifies:

1. the moment recurrence through order 10 at $q = \pm 1/2, \pm 2, \pm 1/4, \pm 4$;
2. negative- q affine conjugacy at every checked moment;
3. reciprocal binomial-convolution inversion through order 10;
4. $q \mapsto q^2$ decimation for $q = \pm 1/2, \pm 2$;
5. the position-energy identity $H_{N,k}(q) = q^{\binom{k}{2}} \begin{bmatrix} N \\ k \end{bmatrix}_q$ for $N \leq 8$;
6. the q -Prouhet transfer for $N \leq 7, \ell \leq 4$, at all eight values;
7. the exact moment-polynomial recurrence and the conjectural odd- q -integer divisor by rational polynomial long division through $n = 16$;
8. inverse-geometric endpoints through $n = 12$ for $q = 1/2, 1/4$;
9. the Grassmannian/unitary normalizations for $Q = 2, 4$ and $N \leq 7$;
10. the reciprocal Hankel signature through order 9 for $q = \pm 2, \pm 4$;
11. the Mahler, sign, reciprocal, and decimation product identities at 90 decimal digits.

The largest high-precision residual is approximately 1.3×10^{-40} ; the remaining analytic residuals are typically around 10^{-90} . The weaker sign-conjugacy residual reflects the deliberately finite 300-factor truncation after an argument dilation as large as five, not a loss of the exact identity.

16.2 Generated files

file	content
<code>analytic_product_errors.csv</code>	high-precision residuals for four analytic identities
<code>q_orbit_moments.csv</code>	exact moments and variances for all eight requested parameters
<code>q_prouhet_verification.csv</code>	exact two-sided values in theorem 12.1
<code>moment_polynomial_divisibility.csv</code>	exact $\mathcal{P}_n$ coefficients, divisors, and quotients through $n = 16$
<code>inverse_geometric_endpoint_values.csv</code>	exact dyadic and quartic endpoint values
<code>grassmannian_unitary_square.csv</code>	positive/negative finite-geometry normalizations
<code>reciprocal_hankel_signatures.csv</code>	exact determinants and predicted signs

The program can be rerun from the report directory with

```
python numerical_experiments.py
```

It creates the `data/` and `assets/` directories if needed, rewrites the CSV files, verifies all assertions, and regenerates the five figures.

16.3 Selected exact moment rows

For reference, the first six raw moments are shown below. They make the reciprocal sign change tangible: positive contractions have positive variance, whereas their reciprocal germs have the opposite centered cumulants.

Table 2: Exact moments $m_n(q)$ for the requested parameter values.

q	m_1	m_2	m_3	m_4	m_5	m_6
$1/2$	$1/2$	$5/18$	$1/6$	$143/1350$	$19/270$	$1153/23814$
$-1/2$	$1/2$	$1/2$	$1/2$	$29/50$	$7/10$	$439/490$
2	$1/2$	$2/9$	$1/12$	$16/675$	$1/270$	$-16/59535$
-2	$1/2$	0	$-1/4$	$-2/25$	$3/10$	$10/49$
$1/4$	$1/2$	$3/10$	$1/5$	$121/850$	$9/85$	$44017/541450$
$-1/4$	$1/2$	$7/18$	$1/3$	$283/918$	$137/459$	$1573771/5262894$
4	$1/2$	$1/5$	$1/20$	$-1/425$	$-1/170$	$121/54145$
-4	$1/2$	$1/9$	$-1/12$	$-31/459$	$49/918$	$213523/2631447$

16.4 Independent checks versus proof

The exact computations were also used as a theorem-discovery filter. Several plausible but false reciprocity formulas for the pole-cleared polynomials $\mathcal{P}_n$ were rejected before inclusion. Only the recurrence, boundaries, and endpoint identity proved in section 9 are stated as theorems. The cyclotomic divisor remains a conjecture despite its symbolic evidence. This separation is important in a subject where $q \leftrightarrow 1/q$ often introduces triangular powers and can make an incorrect formula look valid in low degree.

17 Conjectures and frontier directions

17.1 Natural boundaries in the parameter plane

The centered logarithm has the formal expansion

$$\log C_q(t) = \sum_{m=1}^{\infty} \frac{B_{2m}}{2m} \frac{(1-q)^{2m}}{1-q^{2m}} \frac{t^{2m}}{(2m)!}. \quad (17.1)$$

For fixed nonzero t , the coefficient functions have poles at roots of unity of arbitrarily high order. Some singularities may cancel after exponentiation or analytic continuation, and the point $q = 1$ is a degenerate concentration limit, so the dense-pole heuristic is not by itself a proof.

Conjecture 17.1 (Generic natural boundary). *For generic fixed $t \neq 0$, the unit circle is a natural boundary in the q -plane for $C_q(t)$ after removal of any isolated removable degeneracies. Equivalently, no single-valued meromorphic continuation crosses a nontrivial arc of $|q| = 1$. This is **conjectural/open**.*

A viable proof strategy is to isolate primitive root-of-unity singularities in one cumulant order whose principal part cannot be cancelled by lower orders, then control the exponential remainder. The two-nome partition function may supply finite approximants with explicit root-of-unity zero orders.

17.2 Arithmetic of the moment polynomials

The polynomials $\mathcal{P}_n(q)$ connect moments, Bernoulli limits, q -Pochhammer denominators, and exact endpoint values. Besides theorem 9.3, several questions are immediate.

Open problem 17.2 (Coefficient signs and unimodality). *After dividing $\mathcal{P}_n$ by the conjectural odd- q -integer factor, do the remaining coefficients have a stable sign pattern, log-concavity, or unimodality? The first factorizations suggest that parity and Bernoulli signs must be separated before any positivity statement is formulated.*

Open problem 17.3 (Root geometry). *Describe the zeros of $\mathcal{P}_n$. Which are forced roots of unity, which lie on reciprocal curves, and do the noncyclotomic zeros approach a limiting set? Is there a transform relating them to the vertical zeros of the reciprocal orthogonal polynomials in [section 7](#)?*

Open problem 17.4 (Denominator theorem). *Find the least integer D_n such that $D_n \mathcal{P}_n(q) \in \mathbb{Z}[q]$. Compare its prime valuations with the von Staudt–Clausen denominators of B_{2m} and the Mersenne/Fermat-type factors in $(q; q)_n$ at rational q .*

17.3 The reciprocal orthogonal-polynomial problem

The positive measure L_r is an infinite convolution of Laplace laws at the double geometric/harmonic scales

$$\frac{|1 - r||r|^j}{2\pi k}. \quad (17.2)$$

It is symmetric, analytic, and infinitely divisible. Its orthogonal polynomials are therefore a well-posed new family. Important questions include:

1. determine the Jacobi coefficients $\beta_n^{(r)}$ exactly or asymptotically;
2. find a q -difference or integro-difference equation induced by adding one geometric layer of Laplace variables;
3. determine the limiting zero distribution after the natural scaling;
4. compare $r = 1/2$ and $r = 1/4$ through scale multisection;
5. identify whether the laws belong to a tractable subclass of generalized gamma convolutions and exploit the corresponding Thorin measure.

The last point should be handled carefully: individual Laplace variables are differences of exponentials, but the signed difference structure is not itself a positive gamma convolution on $\mathbb{R}_+$.

17.4 A two-nome asymptotic phase diagram

The master function $\mathcal{Z}_N(u, t; \rho, \sigma)$ invites asymptotics in which $N \rightarrow \infty$ while k/N , m/N , or the nomes vary. Possible regimes include:

- $|\rho| < 1$, $|\sigma| < 1$: convergent position and projection energies;
- $\rho > 1$, $\sigma > 1$: large-deviation concentration on terminal digits;
- ρ or σ on the unit circle: resonance and partial Prouhet cancellation;
- reciprocal pairs $(\rho, \sigma) \mapsto (1/\rho, 1/\sigma)$: digit reversal and dilation;
- negative pairs: parity splitting and affine translation of the projected comb.

A saddle-point analysis of the mixed coefficients $\mathcal{M}_{N,k,m}$ could connect Gaussian-binomial limit shapes to Thue–Morse rarefied sums. This appears to be a genuinely two-dimensional problem; specializing $\rho = \sigma$ too early loses the phase diagram.

17.5 Quartic Cantor renormalization

At $q = 1/4$, finite derivative knots lie on a Cantor skeleton but the limiting density is C^∞ on a full interval. The exact dyadic–quartic factorization provides a renormalization relation between a full grid and two sparse grids. Promising problems are:

1. quantify how the Hausdorff geometry of the knots controls finite-level derivative norms;
2. determine the best uniform and Sobolev convergence rates of quartic box splines to the limiting density;

3. derive a wavelet or multiresolution basis adapted to the digit set $\{0, 1\}$ in base four;
4. compare the zero multiplicity divisor $1 + \lfloor \nu_2(n)/2 \rfloor$ with the dyadic divisor $1 + \nu_2(n)$ under the convolution (6.9);
5. use the exact endpoint lattice to calibrate numerical inverse-CDF algorithms near zero.

18 Lean formalization roadmap

The existing formal development already contains the generic weighted-uniform series, geometric weights, probability law, affine self-similarity, reflection, support, absolute continuity, and the classical Fabius bridge. The results of this report can therefore be added incrementally rather than by introducing a parallel foundation.

18.1 Low-risk algebraic tranche

The following statements require only finite sums, formal power series, or existing probability transport lemmas.

1. **Negative affine conjugacy.** Center each uniform coordinate, use invariance under coordinatewise reflection, and push forward by the affine map.
2. **Geometric multisection.** Reindex the series by $n = mk + r$ and apply independence of coordinate subsequences.
3. **Moment recurrence and reciprocity.** Define a formal or analytic germ interface and prove coefficient uniqueness away from roots of unity.
4. **Position-energy layer.** Reuse the finite q -binomial core and prove the subset-energy expansion directly.
5. **Two-nome partition function.** A finite-product definition and extensionality proof should make multisection nearly automatic.
6. **q -Prouhet transfer.** Factor each exponential difference and compare coefficients in a finite exponential generating function.

These modules would have small analytic dependencies and should compile quickly under the repository's single-worker build policy.

18.2 Endpoint-analysis tranche

The endpoint theorem needs the finite-dimensional uniform-simplex volume. A formal proof can proceed by an affine change of variables from the standard simplex

$$\{u_j \geq 0 : \sum u_j \leq h\}$$

and a lemma showing that the upper cube constraints are inactive under (14.2). Once (14.3) is formalized, reflection gives the moment formula, and the derivative jet follows from the existing CDF density equation. The two-term asymptotic needs only Stirling, moment bounds, and monotonic bracketing; it is far lighter than the existing all-orders saddle machinery.

Suggested files are:

module	principal content
<code>GeometricUniformSign.lean</code>	negative affine law and support endpoints
<code>GeometricUniformMultisection.lean</code>	$q \mapsto q^m$ independent decomposition
<code>GeometricProuhet.lean</code>	finite transfer and normalized moment limit
<code>TwoNomePartition.lean</code>	master product and mixed coefficients
<code>GeometricEndpoint.lean</code>	stop-loss formula and exact lattice
<code>GeometricEndpointJets.lean</code>	derivative hierarchy
<code>GeometricEndpointAsymptotic.lean</code>	two-term CDF and inverse laws

18.3 Reciprocal analytic tranche

The reciprocal Laplace theorem requires more infrastructure:

1. define the reciprocal germ as a locally normally convergent inverse product;
2. prove the Euler product rearrangement;
3. construct the double-indexed Laplace series in L^2 ;
4. identify its characteristic function by finite partial products and dominated convergence;
5. establish strict positivity of real Hankel Gram determinants;
6. transport determinants under affine complex maps;
7. define the induced quasi-definite moment functional and orthogonal polynomial recurrence.

A pragmatic split is to formalize reciprocity and cumulants first, then the Laplace representation, and only then the Hankel/orthogonality consequences.

18.4 Status separation in documentation

The report's theorem/conjecture boundary should be preserved if material is merged into the repository. In particular, the following must remain in the frontier quarantine until proved:

- odd- q -integer divisibility of $\mathcal{P}_n$;
- generic natural boundary at $|q| = 1$;
- all-orders Lambert- W_{-1} expansion for general q ;
- asymptotic statements about reciprocal Jacobi coefficients;
- coefficient positivity or root-location patterns for $\mathcal{P}_n$.

19 Conclusion

The main structural lesson is that “other q values” do not merely replace $1/2$ by another number in an existing formula. Three transformations act on three distinct layers:

- $q \mapsto -q$ changes the outer affine normalization while leaving the centered Rvachev law invariant in distribution;
- $q \mapsto 1/q$ turns geometric smoothing into deconvolution, zeros into poles, and real probability moments into vertical-line complex moments;
- $q \mapsto q^m$ separates scales into congruence classes and connects the dyadic and quartic systems by exact convolution.

The finite Gaussian nome and the geometric projection need not coincide. The two-nome partition function shows that q -Pochhammer coefficients and Thue–Morse Prouhet moments are complementary statistics of the same binary cube. The normalized residual of its maximal Prouhet zero is exactly the q -Fabius MGF.

For the requested values, the picture is now compact. The orbit $\{\pm 1/2, \pm 2\}$ is controlled by the classical Fabius law and finite geometry over $\mathbb{F}_2/\mathbb{F}_4$; the orbit $\{\pm 1/4, \pm 4\}$ is controlled by a quartic smooth law whose finite derivatives sit on a dimension-1/2 Cantor skeleton and whose Gaussian coefficients correspond to $\mathbb{F}_4/\mathbb{F}_{16}$ geometry. Each positive contraction has an exact inverse-geometric endpoint lattice, each negative contraction inherits it by affine transport, and each reciprocal germ has a positive Laplace spectral dual.

The strongest immediately actionable next steps are the Lean formalization of negative affine conjugacy, multisection, the finite q -Prouhet transfer, and the endpoint simplex theorem. The most mathematically open directions are the root-of-unity arithmetic of $\mathcal{P}_n$, the reciprocal vertical-line orthogonal polynomials, and the all-orders quartic/general- q Lambert- W endpoint expansion.

A Detailed identities for the requested parameter values

A.1 Orbit transform table

Let $M = M_{1/2}$ and $N = M_{1/4}$. Then

parameter	exact germ in terms of the positive contraction
$1/2$	$M_{1/2}(t) = M(t)$
$-1/2$	$M_{-1/2}(t) = e^{-t}M(3t)$
2	$M_2(t) = e^t/M(t)$
-2	$M_{-2}(t) = e^{2t}/M(3t)$
$1/4$	$M_{1/4}(t) = N(t)$
$-1/4$	$M_{-1/4}(t) = e^{-t/3}N(5t/3)$
4	$M_4(t) = e^t/N(t)$
-4	$M_{-4}(t) = e^{4t/3}/N(5t/3)$

The centered transforms remove the exponential factors:

$$\begin{aligned} C_{-1/2}(t) &= C_{1/2}(3t), & C_2(t) &= C_{1/2}(t)^{-1}, & C_{-2}(t) &= C_{1/2}(3t)^{-1}, \\ C_{-1/4}(t) &= C_{1/4}(5t/3), & C_4(t) &= C_{1/4}(t)^{-1}, & C_{-4}(t) &= C_{1/4}(5t/3)^{-1}. \end{aligned}$$

A.2 Cumulant scaling table

For $m \geq 1$, let $K_{2m}^{(2)} = \kappa_{2m}(1/2)$ and $K_{2m}^{(4)} = \kappa_{2m}(1/4)$. Then

orbit	q	$-q$	$1/q$	$-1/q$
half	$K_{2m}^{(2)}$	$3^{2m} K_{2m}^{(2)}$	$-K_{2m}^{(2)}$	$-3^{2m} K_{2m}^{(2)}$
quarter	$K_{2m}^{(4)}$	$(5/3)^{2m} K_{2m}^{(4)}$	$-K_{2m}^{(4)}$	$-(5/3)^{2m} K_{2m}^{(4)}$

This table follows at once from affine scaling and reciprocal inversion. It is often more efficient than re-evaluating the rational formula (3.2) eight times.

B Proof details for finite-comb transport

B.1 Negative- q translation

Let $q = -a$ and write

$$A_{-a,N}(\epsilon) = \sum_{j \text{ even}} \epsilon_j a^j - \sum_{j \text{ odd}} \epsilon_j a^j.$$

Define ϵ' by preserving even digits and complementing odd digits. Then

$$\begin{aligned} A_{a,N}(\epsilon') &= \sum_{j \text{ even}} \epsilon_j a^j + \sum_{j \text{ odd}} (1 - \epsilon_j) a^j \\ &= A_{-a,N}(\epsilon) + c_{a,N}. \end{aligned}$$

If there are $m = \lfloor N/2 \rfloor$ odd positions, then $(-1)^{|\epsilon|} = (-1)^m (-1)^{|\epsilon'|}$. Summing the atoms proves (12.11).

B.2 Reciprocal dilation

Reverse the digits, $\epsilon'_j = \epsilon_{N-1-j}$. Then

$$\begin{aligned} A_{q^{-1},N}(\epsilon) &= \sum_{j=0}^{N-1} \epsilon_j q^{-j} \\ &= q^{-(N-1)} \sum_{j=0}^{N-1} \epsilon_{N-1-j} q^j = q^{-(N-1)} A_{q,N}(\epsilon'). \end{aligned}$$

Digit reversal preserves Hamming parity, proving (12.12).

C Moment-polynomial data and the divisor conjecture

The first additional factorizations are

$$\mathcal{P}_5(q) = -\frac{[3]_q[5]_q(q^3 - 2q^2 - 2)}{1440}, \quad (\text{C.1})$$

$$\mathcal{P}_6(q) = \frac{[3]_q[5]_q}{60480} (2q^9 - 6q^8 - 3q^7 + 4q^6 + 6q^5 + 15q^4 + 3q^3 + 24q^2 + 6q + 12), \quad (\text{C.2})$$

$$\mathcal{P}_7(q) = \frac{[3]_q[5]_q[7]_q}{120960} (2q^8 - 6q^7 + 4q^6 - 3q^5 + 6q^4 - 4q^3 + 7q^2 + 3). \quad (\text{C.3})$$

Here $[d]_q = (q^d - 1)/(q - 1)$. These formulas were obtained from the proved recurrence (9.3); they are exact. The extrapolation to all odd $d \leq n$ is the conjectural step.

A root-of-unity proof would need to establish the correct multiplicity of each cyclotomic polynomial. Since

$$\prod_{\substack{3 \leq d \leq n \\ d \text{ odd}}} [d]_q = \prod_{\substack{3 \leq e \leq n \\ e \text{ odd}}} \Phi_e(q)^{\lfloor (n/e + 1)/2 \rfloor},$$

the exponent of Φ_e counts the odd multiples of e not exceeding n . Thus simple vanishing of $\mathcal{P}_n$ at primitive roots is not enough. One must control repeated jets. This is where the finite cancellation-order proposition 10.3 may become useful.

D Reproducibility notes

The script requires Python 3.10 or newer and the packages `numpy`, `matplotlib`, and `mpmath`. Exact arithmetic uses only the standard library. The figures are deterministic:

- densities are Fourier inversions of truncated exact sinc products on a periodic grid of size 2^{16} ;
- negative densities are produced by the exact affine change of variables;
- comb figures enumerate all 2^9 binary vectors;
- endpoint decay uses exact rational moments before conversion to the plotted logarithm.

The source is extensively commented and each CSV row contains enough input parameters to reproduce the underlying assertion independently.

E Bibliographic and corpus references

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